

Notes on Pseudo-Differential Operator

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0 Preliminary

Notation. Special notations we want to emphasize:

- (i) We will refer D_j to be $D_j = -i\partial_j$.
- (ii) The most general form of linear partial differential equation on $\mathbb{R}^n$ of order m is

$$\sum_{\alpha_1 + \dots + \alpha_n \leq m} a_{\alpha_1, \dots, \alpha_n}(x) D_1^{\alpha_1} \dots D_n^{\alpha_n}$$

where $\alpha_1, \dots, \alpha_n$ are non-negative integers and $a_{\alpha_1, \dots, \alpha_n}(x) \in C^\infty(\mathbb{R}^n)$.

- (iii) For simplicity, we will denote $\alpha = (\alpha_1, \dots, \alpha_n)$ and $D^\alpha = D_1^{\alpha_1} \dots D_n^{\alpha_n}$ so the general form of linear partial differential equation is given by

$$\sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha$$

where $|\alpha| = \sum_{j=1}^n \alpha_j$.

- (iv) Leibnitz formula:

$$D^\alpha(fg) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} (D^\beta f)(D^{\alpha-\beta} g)$$

Theorem 0.1. There exists a function $\psi \in C_0^\infty(\mathbb{R}^n)$ such that

- (i) $\psi \geq 0$
- (ii) $\psi(0) \neq 0$
- (iii) $\text{supp}(\psi) \subset B_1(0) := \{x \in \mathbb{R}^n : |x| < 1\}$
- (iv) We have

$$\int_{\mathbb{R}^n} \psi(x) dx = 1$$

Proof. We note that the function

$$\chi(t) = \begin{cases} 0 & \text{if } t \geq 0 \\ e^{-1/t} & \text{if } t < 0 \end{cases}$$

is smooth, i.e. $\chi \in C^\infty(\mathbb{R})$. Moreover, $\chi \geq 0$ and $\chi(1) \neq 0$. We define $\psi_0(x) = \chi(1 - 2|x|^2)$. Since the map $x \mapsto |x|^2$ is smooth, $\psi_0 \in C^\infty(\mathbb{R}^n)$. We set:

$$\psi(x) = \frac{\psi_0(x)}{\int_{\mathbb{R}^n} \psi_0(x) dx}$$

which satisfies the conditions required. □

Lemma 0.2. Suppose $\Omega \subset \mathbb{R}^n$ is open and $K \subset \Omega$ is compact. Then $d(K, \partial\Omega) > 0$ where

$$d(K, \partial\Omega) := \inf_{x \in K, y \in \partial\Omega} |x - y|.$$

Proof. Since K is compact, $L \subset B_R(0)$ for some $R > 0$. Let $\Omega_R = \Omega \cap B_R(0)$. Ω_R is open and bounded with $K \subset \Omega_R$. It is enough to show that $d(K, \partial\Omega_R) > 0$. Since Ω_R is bounded, $\partial\Omega_R$ is compact. Thus the map

$$\begin{aligned} f : K \times \partial\Omega_R &\rightarrow \mathbb{R}_{\geq 0} \\ (x, y) &\mapsto |x - y| \end{aligned}$$

is a continuous map on a compact set, hence it achieves its minimum d at (x_0, y_0) . Suppose that $d = 0$, then $x_0 = y_0$, but $x_0 \in K \subset \Omega_R$ and $y_0 \in \partial\Omega_R \subset \Omega_R^c$ a contradiction. Thus $d > 0$. $\square$

Lemma 0.3. For any $z \in \mathbb{R}^n$ we set $\tau_z f(x) = f(x - z)$. Suppose $p \in [1, \infty)$ and $g \in L^p(\mathbb{R}^n)$. Let $\{z_j\}_{j=1}^\infty \subset \mathbb{R}^n$ be a sequence of points such that $z_j \rightarrow 0$ as $j \rightarrow \infty$. Then:

$$\|\tau_{z_j} g - g\|_{L^p(\mathbb{R}^n)} \rightarrow 0$$

Proof. 1. First, suppose $g = \mathbf{1}_R$, where $R = (a_1, b_1] \times (a_2, b_2] \times \cdots \times (a_n, b_n]$ is a rectangle, with side-lengths $I_m = b_m - a_m$ for $m = 1, \dots, n$. Now, since when a box is translated by a vector z_j each side is translated by a distance of at most $|z_j|$, and has area at most $I_{\max}^{n-1}$. Thus

$$\|\tau_{z_j} g - g\|_{L^p(\mathbb{R}^n)}^p \leq 2n|z_j| I_{\max}^{n-1}$$

We conclude that

$$\lim_{j \rightarrow \infty} \|\tau_{z_j} g - g\|_{L^p(\mathbb{R}^n)} = 0$$

2. Now suppose that $g = \mathbf{1}_A$, where A is a measurable set of finite measure. Fix $\varepsilon > 0$, then there exists a compact $K \subset A$ and an open $U \supset A$ such that $|U \setminus K| < \varepsilon^p$. Since U is open, we can write U as a union of open rectangles:

$$U = \bigcup_{\alpha \in \mathcal{A}} R_\alpha$$

Since K is compact, it is covered by a finite subset of these:

$$K \subset \bigcup_{i=1}^N R_i := B$$

Now, note that $K \subset B \subset U$, so the symmetric difference $A \Delta B \subset U \setminus K$. Thus $\|\mathbf{1}_A - \mathbf{1}_B\|_{L^p(\mathbb{R}^n)} = |A \Delta B|^{1/p} < \varepsilon$. By (1) we know that there exists J such that for all $j \geq J$ we have:

$$\|\tau_{z_j} \mathbf{1}_B - \mathbf{1}_B\|_{L^p(\mathbb{R}^n)} < \varepsilon$$

Therefore,

$$\begin{aligned}
\|\tau_{z_j} \mathbf{1}_A - \mathbf{1}_A\|_{L^p(\mathbb{R}^n)} &= \|\tau_{z_j} \mathbf{1}_A - \tau_{z_j} \mathbf{1}_B + \tau_{z_j} \mathbf{1}_B - \mathbf{1}_B + \mathbf{1}_B - \mathbf{1}_A\|_{L^p(\mathbb{R}^n)} \\
&\leq \|\tau_{z_j} \mathbf{1}_A - \tau_{z_j} \mathbf{1}_B\|_{L^p(\mathbb{R}^n)} + \|\tau_{z_j} \mathbf{1}_B - \mathbf{1}_B\|_{L^p(\mathbb{R}^n)} + \|\mathbf{1}_B - \mathbf{1}_A\|_{L^p(\mathbb{R}^n)} \\
&= 2\|\mathbf{1}_B - \mathbf{1}_A\|_{L^p(\mathbb{R}^n)} + \|\tau_{z_j} \mathbf{1}_B - \mathbf{1}_B\|_{L^p(\mathbb{R}^n)} \\
&< 3\varepsilon
\end{aligned}$$

for all $j \geq J$. Thus

$$\lim_{j \rightarrow \infty} \|\tau_{z_j} g - g\|_{L^p(\mathbb{R}^n)} = 0$$

3. Now suppose g is a simple function, $g = \sum_{i=1}^N g_i \mathbf{1}_{A_i}$, where $g_i \in \mathbb{C}$ and A_i are measurable sets of finite measure. Then we have

$$\|\tau_{z_j} g - g\|_{L^p(\mathbb{R}^n)} \leq \sum_{i=1}^N |g_i| \|\tau_{z_j} \mathbf{1}_{A_i} - \mathbf{1}_{A_i}\|_{L^p(\mathbb{R}^n)}$$

so as $j \rightarrow \infty$ we have:

$$\lim_{j \rightarrow \infty} \|\tau_{z_j} g - g\|_{L^p(\mathbb{R}^n)} = 0$$

4. Now suppose that $g \in L^p(\mathbb{R}^n)$. Fix $\varepsilon > 0$, there exists a simple function $\tilde{g} = \sum_{i=1}^n \tilde{g}_i \mathbf{1}_{A_i}$ with $\tilde{g}_i \in \mathbb{C}$, $|A_i| < \infty$ such that $\|g - \tilde{g}\|_{L^p(\mathbb{R}^n)} < \varepsilon$. Then we can find J such that $\|\tau_{z_j} \tilde{g} - \tilde{g}\|_{L^p(\mathbb{R}^n)} < \varepsilon$ for all $j \geq J$. Now:

$$\begin{aligned}
\|\tau_{z_j} g - g\|_{L^p(\mathbb{R}^n)} &= \|\tau_{z_j} g - \tau_{z_j} \tilde{g} + \tau_{z_j} \tilde{g} - \tilde{g} + \tilde{g} - g\|_{L^p(\mathbb{R}^n)} \\
&\leq \|\tau_{z_j} g - \tau_{z_j} \tilde{g}\|_{L^p(\mathbb{R}^n)} + \|\tau_{z_j} \tilde{g} - \tilde{g}\|_{L^p(\mathbb{R}^n)} + \|\tilde{g} - g\|_{L^p(\mathbb{R}^n)} \\
&= 2\|g - \tilde{g}\|_{L^p(\mathbb{R}^n)} + \|\tau_{z_j} \tilde{g} - \tilde{g}\|_{L^p(\mathbb{R}^n)} \\
&< 3\varepsilon
\end{aligned}$$

Thus we conclude that

$$\lim_{j \rightarrow \infty} \|\tau_{z_j} g - g\|_{L^p(\mathbb{R}^n)} = 0$$

□

Definition 0.4. If f, g are functions mapping $\mathbb{R}^n$ to $\mathbb{C}$, then we define the convolution of f and g by

$$(f * g)(x) = \int_{\mathbb{R}^n} f(y)g(x-y)dy$$

if the integral exists.

Lemma 0.5. Suppose $f, g, h \in C_0^\infty(\mathbb{R}^n)$. Then:

$$f * g = g * f, \quad f * (g * h) = (f * g) * h$$

and

$$\int_{\mathbb{R}^n} (f * g)(x)dx = \int_{\mathbb{R}^n} f(x)dx \int_{\mathbb{R}^n} g(x)dx$$

Proof. With the change of variables $y = x - z$, we have

$$(f * g)(x) = \int_{\mathbb{R}^n} f(y)g(x-y)dy = \int_{\mathbb{R}^n} f(x-z)g(z)dz = (g * f)(x)$$

Next, we see that

$$\begin{aligned} [f * (g * h)](x) &= \int_{\mathbb{R}^n} f(y) \left(\int_{\mathbb{R}^n} g(z)h(x-y-z)dz \right) dy \\ &= \int_{\mathbb{R}^n} f(y) \left(\int_{\mathbb{R}^n} g(w-y)h(x-w) dw \right) dy \\ &= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} f(y)g(w-y)dy \right) h(x-w) dw \\ &= [(f * g) * h](x) \end{aligned}$$

Note that Fubini's theorem is applicable since $f, g, h \in C_0^\infty(\mathbb{R}^n)$. Finally, we calculate:

$$\begin{aligned} \int_{\mathbb{R}^n} (f * g)(x)dx &= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} f(y)g(x-y)dy \right) dx \\ &= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} f(y)g(x-y)dx \right) dy \\ &= \int_{\mathbb{R}^n} (f(y) \int_{\mathbb{R}^n} g(z)dz)dy \\ &= \int_{\mathbb{R}^n} f(x)dx \int_{\mathbb{R}^n} g(x)dx \end{aligned}$$

□

Lemma 0.6. We define the difference quotient by

$$\Delta_i^h f(x) = \frac{f(x + he_i) - f(x)}{h}$$

then

- (1) Suppose $f \in C_0^0(\mathbb{R}^n)$ and $\{z_i\}_{i \in \mathbb{N}} \subset \mathbb{R}^n$ is a sequence with $z_i \rightarrow 0$ as $i \rightarrow \infty$. Then for any $x \in \mathbb{R}^n$:
 - (i) $\tau_{z_j} f(x) \rightarrow f(x)$ as $j \rightarrow \infty$.
 - (ii) $|\tau_{z_j} f(x)| \leq (\sup_{\mathbb{R}^n} |f|) \mathbf{1}_{B_R(0)}(x)$ for some $R > 0$ and all j .
- (2) Suppose $f \in C_0^1(\mathbb{R}^n)$ and $\{h_j\}_{j \in \mathbb{N}} \subset \mathbb{R}$ is a sequence with $h_j \rightarrow 0$ as $j \rightarrow \infty$. Then for any $x \in \mathbb{R}^n$:
 - (i) $\Delta_i^{h_j} f(x) \rightarrow \partial_i f(x)$ as $j \rightarrow \infty$.
 - (ii) $|\Delta_i^{h_j} f(x)| \leq (\sup_{\mathbb{R}^n} |\partial_i f|) \mathbf{1}_{B_R(0)}(x)$ for some $R > 0$ and all j .

Proof.

- (1) (i) Recall that $\tau_{z_j} f(x) = f(x - z_j)$. Clearly since $z_j \rightarrow 0$, $f(x - z_j) \rightarrow f(x)$ as $j \rightarrow \infty$ due to the continuity of f .
(ii) Since $z_j \rightarrow 0$, there exists some $\rho > 0$ such that $z_j \in \overline{B_\rho(0)}$ for all j .
Now

$$\text{supp}(\tau_{z_j} f) = \text{supp}(f) + z_j \subset \text{supp}(f) + \overline{B_\rho(0)}$$

Since the sum of two bounded set is bounded, we conclude that there exists $R > 0$ such that $\text{supp}(\tau_{z_j} f) \subset B_R(0)$. Thus $\tau_{z_j} f = \tau_{z_j} f \mathbf{1}_{B_R(0)}$ and we estimate:

$$|\tau_{z_j} f(x)| = |\tau_{z_j} f(x)| \mathbf{1}_{B_R(0)}(x) \leq \sup_{\mathbb{R}^n} |f| \mathbf{1}_{B_R(0)}(x).$$

- (2) Suppose that $f \in C_0^1(\mathbb{R}^n)$ and $\{h_j\}_{j \in \mathbb{N}} \subset \mathbb{R}$ is a sequence with $h_j \rightarrow 0$ as $j \rightarrow \infty$. Then for any $x \in \mathbb{R}^n$:

- (i) From the definition of the difference quotient and of the partial derivative:

$$\Delta_i^{h_j} f(x) = \frac{f(x + h_j e_i) - f(x)}{h_j} \rightarrow \partial_i f(x), \quad \text{as } j \rightarrow \infty$$

- (ii) Since $h_j \rightarrow 0$, there is some $\rho > 0$ such that $|h_j| \leq \rho$ for all j . We have

$$\begin{aligned} \text{supp}(\Delta_i^{h_j} f) &\subset \text{supp}(\tau_{-h_j e_i} f) \cup \text{supp}(f) = (\text{supp}(f) - h_j e_i) \cup \text{supp}(f) \\ &\subset (\text{supp}(f) + \overline{B_\rho(0)}) \cup \text{supp}(f) \subset B_R(0) \end{aligned}$$

for some $R > 0$ since the union of two bounded sets is bounded. Thus $\Delta_i^{h_j} f = \Delta_i^{h_j} f \mathbf{1}_{B_R(0)}$. In addition, by the mean value theorem, for any $h \in \mathbb{R}$, there exists $s \in \mathbb{R}$ with $|s| < |h|$ such that

$$\frac{f(x + h e_i) - f(x)}{h} = \partial_i f(x + s e_i)$$

thus

$$|\Delta_i^{h_j} f(x)| \leq \sup_{\mathbb{R}^n} |\partial_i f|$$

Therefore,

$$|\Delta_i^{h_j} f(x)| = |\Delta_i^{h_j} f(x)| \mathbf{1}_{B_R(0)}(x) \leq \sup_{\mathbb{R}^n} |\partial_i f| \mathbf{1}_{B_R(0)}(x).$$

□

Theorem 0.7. Suppose $f \in L_{loc}^1(\mathbb{R}^n)$ and $g \in C_c^k(\mathbb{R}^n)$ for some $k \geq 0$. Then $f * g \in C^k(\mathbb{R}^n)$ and

$$\partial^\alpha (f * g) = f * \partial^\alpha g$$

for any multi-index with $|\alpha| \leq k$.

Proof. We first establish the result for $k = 0$. We need to show that if $f \in L^1_{loc}(\mathbb{R}^n)$ and $g \in C^0_0(\mathbb{R}^n)$ then $f * g$ is continuous. To show this, it suffices to show that $f * g(x - z_j) \rightarrow f * g(x)$ for any $x \in \mathbb{R}^n$ and any sequence $\{z_j\}_{j \in \mathbb{N}}$ with $z_j \rightarrow 0$. Now, note that

$$f * g(x - z_j) = \int_{\mathbb{R}^n} f(y)g(x - z_j - y)dy = \int_{\mathbb{R}^n} f(y)\tau_{z_j}g(x - y)dy$$

Note that for any fixed x and all j :

$$|f(y)\tau_{z_j}g(x - y)| \leq \sup_{\mathbb{R}^n} |g|\mathbf{1}_{B_R(0)}(x - y)|f(y)|$$

for some R by previous Lemma. Since $f \in L^1_{loc}(\mathbb{R}^n)$ the RHS is integrable, and so by the DCT

$$\lim_{j \rightarrow \infty} f * g(x - z_j) = \int_{\mathbb{R}^n} \lim_{j \rightarrow \infty} f(y)\tau_{z_j}g(x - y)dy = \int_{\mathbb{R}^n} f(y)g(x - y)dy = f * g(x)$$

Now suppose that $f \in L^1_{loc}(\mathbb{R}^n)$ and $g \in C^1_0(\mathbb{R}^n)$. Clearly $f * \partial_i g$ is continuous by the previous argument. To show $f * g \in C^1(\mathbb{R}^n)$, it suffices to show that for any $x \in \mathbb{R}^n$ and any sequence $\{h_j\}_{j \in \mathbb{N}} \subset \mathbb{R}$ with $h_j \rightarrow 0$ we have:

$$\lim_{j \rightarrow \infty} \Delta_i^{h_j} f * g(x) = f * \partial_i g(x)$$

Note that

$$\begin{aligned} \Delta_i^{h_j} f * g(x) &= \frac{f * g(x + h_j e_i) - f * g(x)}{h_j} \\ &= \int_{\mathbb{R}^n} f(y) \left(\frac{g(x + h_j e_i - y) - g(x - y)}{h_j} \right) dy \\ &= \int_{\mathbb{R}^n} f(y) \Delta_i^{h_j} g(x - y) dy \end{aligned}$$

Again by previous Lemma we see that the DCT is applicable thus

$$\lim_{j \rightarrow \infty} \Delta_i^{h_j} f * g(x) = \int_{\mathbb{R}^n} \lim_{j \rightarrow \infty} f(y) \Delta_i^{h_j} g(x - y) dy = f * \partial_i g(x).$$

The case where $g \in C^k_0(\mathbb{R}^n)$ with $k > 1$ follows by induction. $\square$

Lemma 0.8. Suppose $f \in L^1_{loc}(\mathbb{R}^n)$ and $g \in C^k_0(\mathbb{R}^n)$ for some $k \geq 0$. Then

$$\text{supp}(f * g) \subset \text{supp}(f) + \text{supp}(g)$$

Proof. Recall that

$$f * g(x) = \int_{\mathbb{R}^n} f(y)g(x - y)dy$$

Thus, if $f * g(x) \neq 0$, then there must exist $y \in \mathbb{R}^n$ such that $y \in \text{supp}(f)$ and $x - y = z \in \text{supp}(g)$. Thus $x = y + z$ with $y \in \text{supp}(f)$ and $z \in \text{supp}(g)$. This means

$$\{x \in \mathbb{R}^n : f * g(x) \neq 0\} \subset \text{supp}(f) + \text{supp}(g)$$

Since $\text{supp}(f)$ is closed and $\text{supp}(g)$ is compact, we know that $\text{supp}(f) + \text{supp}(g)$ is closed, thus

$$\text{supp}(f * g) = \overline{\{x \in \mathbb{R}^n : f * g(x) \neq 0\}} \subset \text{supp}(f) + \text{supp}(g).$$

as required. $\square$

Theorem 0.9. Suppose $\phi \in C_0^\infty(\mathbb{R}^n)$ satisfies:

- (i) $\phi \geq 0$
- (ii) $\text{supp}(\phi) \subset B_1(0)$
- (iii) $\int_{\mathbb{R}^n} \phi(x) dx = 1$

such ϕ exists by Theorem 0.1. Define:

$$\phi_\varepsilon(y) = \frac{1}{\varepsilon^n} \phi\left(\frac{y}{\varepsilon}\right)$$

Then:

- (1) If $f \in C_0^k(\mathbb{R}^n)$, then $\phi_\varepsilon * f$ is smooth, and

$$\partial^\alpha(\phi_\varepsilon * f) \rightarrow \partial^\alpha f \quad \text{as } \varepsilon \rightarrow 0$$

uniformly on $\mathbb{R}^n$ for any multi-index with $|\alpha| \leq k$.

- (2) If $g \in L^p(\mathbb{R}^n)$ with $1 \leq p < \infty$, then $\phi_\varepsilon * g$ is smooth, and

$$\phi_\varepsilon * g \rightarrow g \quad \text{in } L^p(\mathbb{R}^n) \quad \text{as } \varepsilon \rightarrow 0$$

- (3) Suppose $f \in C^k(\mathbb{R}^n)$ with $\sup_{\mathbb{R}^n} |\partial^\alpha f| < \infty$ for $|\alpha| \leq k$, and suppose $g \in L^1(\mathbb{R}^n)$ with $g \geq 0$, $\int_{\mathbb{R}^n} g(x) dx = 1$. Set $g_\varepsilon(y) = \varepsilon^{-n} g(\varepsilon^{-1}y)$. Then $f * g_\varepsilon \in C^k(\mathbb{R}^n)$, and

$$\partial^\alpha(f * g_\varepsilon)(x) \rightarrow \partial^\alpha f(x) \quad \text{as } \varepsilon \rightarrow 0$$

for any $x \in \mathbb{R}^n$ and any multi-index with $|\alpha| \leq k$.

Proof.

- (1) Note that ϕ_ε also satisfies

$$\int_{\mathbb{R}^n} \phi_\varepsilon(y) dy = 1$$

By Theorem 0.7, we have $\partial^\alpha(\phi_\varepsilon * f) = \phi_\varepsilon * \partial^\alpha f$ for any $|\alpha| \leq k$. Using these two facts, we calculate:

$$\begin{aligned}\partial^\alpha(\phi_\varepsilon * f)(x) - \partial^\alpha f(x) &= \int_{\mathbb{R}^n} \phi_\varepsilon(y) \partial^\alpha f(x-y) dy - \partial^\alpha f(x) \int_{\mathbb{R}^n} \phi_\varepsilon(y) dy \\ &= \int_{\mathbb{R}^n} \phi_\varepsilon(y) [\partial^\alpha f(x-y) - \partial^\alpha f(x)] dy \\ &= \int_{B_1(0)} \phi(z) [\partial^\alpha f(x - \varepsilon z) - \partial^\alpha f(x)] dz\end{aligned}$$

Now, since $\phi \geq 0$, we can estimate:

$$\begin{aligned}|\partial^\alpha(\phi_\varepsilon * f)(x) - \partial^\alpha f(x)| &\leq \int_{B_1(0)} \phi(z) |\partial^\alpha f(x - \varepsilon z) - \partial^\alpha f(x)| dz \\ &\leq \sup_{z \in B_1(0)} |\partial^\alpha f(x - \varepsilon z) - \partial^\alpha f(x)| \times \int_{B_1(0)} \phi(z) dz \\ &= \sup_{z \in B_1(0)} |\partial^\alpha f(x - \varepsilon z) - \partial^\alpha f(x)|\end{aligned}$$

Now, since $\partial^\alpha f$ is continuous and of compact support, it is uniformly continuous on $\mathbb{R}^n$. Fix $\tilde{\varepsilon} > 0$, there exists δ such that for any $v, w \in \mathbb{R}^n$ with $|x - y| < \delta$, we have

$$|\partial^\alpha f(v) - \partial^\alpha f(w)| < \tilde{\varepsilon}$$

For any $x \in \mathbb{R}^n$, taking $\varepsilon < \delta$, and $v = x - \varepsilon z$, $w = x$ with $z \in B_1(0)$ we have $|v - w| < \delta$, so:

$$|\partial^\alpha f(x - \varepsilon z) - \partial^\alpha f(x)| < \tilde{\varepsilon}$$

holds for any $x \in \mathbb{R}^n$, $z \in B_1(0)$. Thus we have

$$\sup_{x \in \mathbb{R}^n} |\partial^\alpha(\phi_\varepsilon * f)(x) - \partial^\alpha f(x)| < \tilde{\varepsilon}.$$

This is the statement of uniform convergence on $\mathbb{R}^n$.

- (2) Noting that $L^p(\mathbb{R}^n) \subset L^1_{loc}(\mathbb{R}^n)$ by Hölder's inequality. Theorem 0.7 immediately establishes the smoothness of $\phi_\varepsilon * g$. We shall use the Minkowski's Integral identity which states that for $F : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{C}$ a measurable function, we have the estimate:

$$\left[\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) dx \right|^p dy \right]^{1/p} \leq \int_{\mathbb{R}^n} \left[\int_{\mathbb{R}^n} |F(x, y)|^p dy \right]^{1/p} dx$$

Now we already know that

$$|(\phi_\varepsilon * g)(x) - g(x)| \leq \int_{\mathbb{R}^n} \phi(z) |g(x - \varepsilon z) - g(x)| dz$$

Integrating and apply Minkowski's integral inequality we have

$$\begin{aligned}
\|\phi_\varepsilon * g - g\|_{L^p(\mathbb{R}^n)} &= \left[\int_{\mathbb{R}^n} |(\phi_\varepsilon * g)(x) - g(x)|^p dx \right]^{1/p} \\
&\leq \left[\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} \phi(z) |g(x - \varepsilon z) - g(x)| dz \right|^p dx \right]^{1/p} \\
&\leq \int_{\mathbb{R}^n} \left[\int_{\mathbb{R}^n} \phi(z)^p |g(x - \varepsilon z) - g(x)|^p dx \right]^{1/p} dz \\
&= \int_{\mathbb{R}^n} \phi(z) \|\tau_{\varepsilon z} g - g\|_{L^p(\mathbb{R}^n)} dz
\end{aligned}$$

to establish our result it will suffice to set $\varepsilon = \varepsilon_j$, where $\{\varepsilon_j\}_{j \in \mathbb{N}} \subset \mathbb{R}$ is any sequence with $\varepsilon_j \rightarrow 0$, and show that $\|\phi_{\varepsilon_j} * g - g\|_{L^p(\mathbb{R}^n)} \rightarrow 0$. Note that since $\|\tau_{\varepsilon_j z} g\|_{L^p(\mathbb{R}^n)} = \|g\|_{L^p(\mathbb{R}^n)}$ we have

$$\phi(z) \|\tau_{\varepsilon_j z} g - g\|_{L^p(\mathbb{R}^n)} \leq 2\phi(z) \|g\|_{L^p(\mathbb{R}^n)}$$

Now by Lemma 0.3 we see that for each $z \in \mathbb{R}^n$ we have

$$\lim_{j \rightarrow \infty} \|\tau_{\varepsilon_j z} g - g\|_{L^p(\mathbb{R}^n)} = 0$$

Then by the DCT we conclude that

$$\lim_{j \rightarrow \infty} \|\phi_{\varepsilon_j} * g - g\|_{L^p(\mathbb{R}^n)} = 0.$$

- (3) Again, by Theorem 0.1 we have that $\partial^\alpha(f * g_\varepsilon) = \partial^\alpha f * g_\varepsilon$ for any $|\alpha| \leq k$. By a change of variables, we calculate:

$$\partial^\alpha(f * g_\varepsilon)(x) = \int_{\mathbb{R}^n} g_\varepsilon(y) \partial^\alpha f(x - y) dy = \int_{\mathbb{R}^n} g(z) \partial^\alpha f(x - \varepsilon z) dz$$

Clearly for each fixed $x \in \mathbb{R}^n$:

$$g(z) \partial^\alpha f(x - \varepsilon z) \rightarrow g(z) \partial^\alpha f(x)$$

for $z \in \mathbb{R}^n$ as $\varepsilon \rightarrow 0$. In addition,

$$|g(z) \partial^\alpha f(x - \varepsilon z)| \leq g(z) \sup_{\mathbb{R}^n} |\partial^\alpha f|$$

thus by the DCT, we conclude

$$\partial^\alpha(f * g_\varepsilon)(x) \rightarrow \partial^\alpha f(x) \int_{\mathbb{R}^n} g(z) dz = \partial^\alpha f(x)$$

as $\varepsilon \rightarrow 0$.

□

Lemma 0.10. Suppose $\Omega \subset \mathbb{R}^n$ is open, and $K \subset \Omega$ is compact. Then there exists $\eta \in C_0^\infty(\Omega)$ such that $\eta = 1$ in a neighbourhood of K .

Proof. By Lemma 0.7, there exists $\varepsilon > 0$ such that $d(K, \partial\Omega) > 4\varepsilon$. We define $K_\varepsilon = K + \overline{B_{2\varepsilon}(0)}$. As the sum of two compact sets, K_ε is compact. Moreover, $K_\varepsilon \subset \Omega$. Suppose ϕ_ε is as in Theorem 0.9. Consider

$$\eta := \phi_\varepsilon * \mathbf{1}_{K_\varepsilon}$$

Then Theorem 0.1 implies that $\eta \in C^\infty(\mathbb{R}^n)$. And Lemma 0.8 implies that

$$\text{supp}(\eta) = K_\varepsilon + \text{supp}(\phi_\varepsilon) \subset K + \overline{B_{2\varepsilon}(0)} + \overline{B_\varepsilon(0)} = K + \overline{B_{3\varepsilon}(0)} \subset \Omega$$

Thus $\eta \in C_0^\infty(\Omega)$. Now, suppose $x \in K + B_\varepsilon(0)$. Then $x + B_\varepsilon(0) \subset K_\varepsilon$ and so

$$\begin{aligned} \eta(x) &= \int_{\mathbb{R}^n} \phi_\varepsilon(y) \mathbf{1}_{K_\varepsilon}(x - y) dy \\ &= \int_{B_\varepsilon(0)} \phi_\varepsilon(y) \mathbf{1}_{K_\varepsilon}(x - y) dy \\ &= \int_{B_\varepsilon(0)} \phi_\varepsilon(y) dy = 1 \end{aligned}$$

Thus $\eta(x) = 1$ for $x \in K + B_\varepsilon(0)$, which is the neighbourhood of K . $\square$

Definition 0.11. A function $\phi \in C^\infty(\mathbb{R}^n)$ is said to be rapidly decreasing if:

$$\sup_{x \in \mathbb{R}^n} |(1 + |x|)^N \partial^\alpha \phi(x)| < \infty$$

for all multi-indices α and all $N \in \mathbb{N}$.

Definition 0.12. We define a family of semi-norms by $\mathcal{P} = \{p_N\}_{N \in \mathbb{N}}$, where

$$p_N(\phi) = \sup_{x \in \mathbb{R}^n, |\alpha| \leq N} |(1 + |x|)^N \partial^\alpha \phi(x)|.$$

This is a separating family of semi-norms, giving rise to a topology $\tau_{\mathcal{P}}$ such that vector space operations are continuous. We denote the set of rapidly decreasing functions equipped with the topology $\tau_{\mathcal{P}}$ by $\mathcal{S}(\mathbb{R}^n)$ or $\mathcal{S}$. This is known as the Schwartz class of functions.

Definition 0.13. For $f \in L^1(\mathbb{R}^n)$, we define the Fourier transform $\hat{f} : \mathbb{R}^n \rightarrow \mathbb{C}$ by:

$$\mathcal{F}[f](\xi) = \hat{f}(\xi) := (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} dx.$$

this is well defined since $|f(x) e^{-ix \cdot \xi}| \leq |f(x)|$ and $f(x) \in L^1(\mathbb{R}^n)$.

Lemma 0.14 (Riemann-Lebesgue Lemma). Suppose $f \in L^1(\mathbb{R}^n)$. Then $\hat{f} \in C^0(\mathbb{R}^n)$ with the estimate

$$\sup_{\xi \in \mathbb{R}^n} |\hat{f}(\xi)| \leq (2\pi)^{-n/2} \|f\|_{L^1}$$

and moreover $\hat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$.

Proof. To establish the continuity of $\hat{f}$, we use the continuity of the exponential, together with the DCT. Let $\{\xi_j\}_{j \in \mathbb{N}}$ be any sequence with $\xi_j \rightarrow \xi$ as $j \rightarrow \infty$. Recalling that

$$\hat{f}(\xi_j) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi_j} dx.$$

Now, clearly for $x \in \mathbb{R}^n$ we have

$$f(x) e^{-ix \cdot \xi_j} \rightarrow f(x) e^{-ix \cdot \xi}, \text{ as } j \rightarrow \infty$$

so we have pointwise convergence of the integrand. We can also estimate:

$$|f(x) e^{-ix \cdot \xi_j}| \leq |f(x)|$$

Now by applying the DCT, we conclude:

$$\hat{f}(\xi_j) \rightarrow \hat{f}(\xi), \text{ as } j \rightarrow \infty.$$

This implies that $\hat{f}(\xi)$ is continuous. We can estimate:

$$\begin{aligned} \sup_{\xi \in \mathbb{R}^n} |\hat{f}(\xi)| &= \sup_{\xi \in \mathbb{R}^n} |(2\pi)^{-n/2} \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} dx| \\ &\leq (2\pi)^{-n/2} \sup_{\xi \in \mathbb{R}^n} \int_{\mathbb{R}^n} |f(x)| dx \\ &= (2\pi)^{-n/2} \|f\|_{L^1} \end{aligned}$$

We see that if $f \in L^1(\mathbb{R}^n)$, we can approximate f by an element of $C_0^\infty(\mathbb{R}^n)$ by Theorem 0.9. Given $\varepsilon > 0$, there exists $f_\varepsilon \in C_0^\infty(\mathbb{R}^n)$ with

$$\|f - f_\varepsilon\|_{L^1} < \frac{\varepsilon}{2}.$$

Now, in the integral for $\hat{f}_\varepsilon$ we can integrate by parts:

$$\begin{aligned} \hat{f}_\varepsilon(\xi) &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} f_\varepsilon(x) e^{-ix \cdot \xi} dx \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} f_\varepsilon(x) \operatorname{div} \left(\frac{\xi}{-i|\xi|^2} e^{-ix \cdot \xi} \right) dx \\ &= -(2\pi)^{-n/2} \int_{\mathbb{R}^n} \frac{\xi}{-i|\xi|^2} \cdot \partial f_\varepsilon(x) e^{-ix \cdot \xi} dx \end{aligned}$$

so that for each $i = 1, \dots, n$ we have, by the Cauchy-Schwarz inequality:

$$\begin{aligned} |\hat{f}_\varepsilon(\xi)| &= \left| (2\pi)^{-n/2} \int_{\mathbb{R}^n} \frac{\xi}{-i|\xi|^2} \cdot \partial f_\varepsilon(x) e^{-ix \cdot \xi} dx \right| \\ &\leq (2\pi)^{-n/2} \int_{\mathbb{R}^n} \left| \frac{\xi}{-i|\xi|^2} \cdot \partial f_\varepsilon(x) e^{-ix \cdot \xi} \right| dx \\ &\leq (2\pi)^{-n/2} \int_{\mathbb{R}^n} \frac{1}{|\xi|} |\partial f_\varepsilon(x)| dx \\ &= \frac{(2\pi)^{-n/2}}{|\xi|} \|\partial f_\varepsilon(x)\|_{L^1} \end{aligned}$$

From this, we conclude that there exists $R > 0$ such that if $|\xi| > R$, we have $|\hat{f}_\varepsilon(\xi)| < \frac{\varepsilon}{2}$. For $|\xi| > R$ we calculate:

$$\begin{aligned} |\hat{f}(\xi)| &= |\hat{f}(\xi) - \hat{f}_\varepsilon(\xi) + \hat{f}_\varepsilon(\xi)| \\ &\leq |\hat{f}_\varepsilon(\xi)| + |\hat{f}(\xi) - \hat{f}_\varepsilon(\xi)| \\ &\leq |\hat{f}_\varepsilon(\xi)| + \|f - f_\varepsilon\|_{L^1} < \varepsilon. \end{aligned}$$

Since ε was arbitrary, we have shown that $|\hat{f}(\xi)| \rightarrow 0$. $\square$

Lemma 0.15. Recall that $\tau_y f(x) = f(x - y)$ and introduce the character $e_y(x) = e^{iy \cdot x}$. Then

- (1) Suppose $f \in L^1(\mathbb{R}^n)$, $x \in \mathbb{R}^n$, $\lambda > 0$ and $f_\lambda(y) = \lambda^{-n} f(\lambda^{-1}y)$. Then

$$\hat{f}_\lambda(\xi) = \hat{f}(\lambda\xi), \quad (\widehat{e_x f})(\xi) = \tau_x \hat{f}(\xi), \quad \widehat{\tau_x f}(\xi) = e_{-x}(\xi) \hat{f}(\xi).$$

- (2) Suppose $f, g \in L^1(\mathbb{R}^n)$. Then $f * g \in L^1(\mathbb{R}^n)$ and

$$\widehat{f * g}(\xi) = (2\pi)^{n/2} \hat{f}(\xi) \hat{g}(\xi).$$

Proof.

- (1) By definition and the change of variable to $z = \lambda^{-1}x$ we have

$$\begin{aligned} \hat{f}_\lambda(\xi) &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} f_\lambda(x) e^{-i\xi \cdot x} dx = (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(\lambda^{-1}x) e^{-i\xi \cdot x} \lambda^{-n} dx \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(y) e^{-i\lambda\xi \cdot z} dz = \hat{f}(\lambda\xi) \end{aligned}$$

Next, we calculate:

$$\begin{aligned} (\widehat{e_x f})(\xi) &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot y} f(y) e^{-i\xi \cdot y} dy \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(y) e^{-i(\xi - x) \cdot y} dy \\ &= \tau_x \hat{f}(\xi) \end{aligned}$$

Finally, we have:

$$\begin{aligned} \widehat{\tau_x f}(\xi) &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(y - x) e^{-i\xi \cdot y} dy = (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(z) e^{-i\xi \cdot (z+x)} dz \\ &= e^{-i\xi \cdot x} (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(z) e^{-i\xi \cdot z} dz = e_{-x}(\xi) \hat{f}(\xi) \end{aligned}$$

where we have used the substitution $z = y - x$.

(2) We first show that $f * g \in L^1(\mathbb{R}^n)$. To see this note that

$$|f * g(x)| = \left| \int_{\mathbb{R}^n} f(y)g(x-y)dy \right| \leq \int_{\mathbb{R}^n} |f(y)g(x-y)|dy$$

Integrating and applying Tonelli's theorem, we have:

$$\begin{aligned} \|f * g\|_{L^1} &\leq \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |f(y)g(x-y)|dy \right) dx \\ &= \int_{\mathbb{R}^n} |f(y)| \left(\int_{\mathbb{R}^n} |g(x-y)|dx \right) dy \\ &= \int_{\mathbb{R}^n} |f(y)| \|g\|_{L^1} dy = \|f\|_{L^1} \|g\|_{L^1} \end{aligned}$$

Now, we can calculate the Fourier transform:

$$\begin{aligned} \widehat{f * g}(\xi) &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} f * g(x) e^{-i\xi \cdot x} dx \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} f(y)g(x-y)dy \right) e^{-i\xi \cdot x} dx \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(y) \left(\int_{\mathbb{R}^n} g(x-y) e^{-i\xi \cdot x} dx \right) dy \\ &= \int_{\mathbb{R}^n} f(y) \widehat{\tau_y g}(\xi) dy \\ &= \int_{\mathbb{R}^n} f(y) \hat{g}(\xi) e^{-i\xi \cdot y} dy \\ &= (2\pi)^{n/2} \hat{f}(\xi) \hat{g}(\xi) \end{aligned}$$

□

Theorem 0.16.

(1) Suppose $f \in C^1(\mathbb{R}^n)$ and that $f, \partial_j f \in L^1(\mathbb{R}^n)$ for all $j = 1, \dots, n$. Then

$$\widehat{\partial_j f}(\xi) = i\xi_j \hat{f}(\xi)$$

(2) Suppose $(1 + |x|)f \in L^1(\mathbb{R}^n)$. Then $\hat{f} \in C^1(\mathbb{R}^n)$, and:

$$\partial_j \hat{f}(\xi) = -i\xi_j \widehat{xf}(\xi)$$

Proof.

(1) For $f \in C^1(\mathbb{R}^n)$ with $f, \partial_i f \in L^1(\mathbb{R}^n)$, then for any $\varepsilon > 0$ there exists $f_\varepsilon \in C_0^1(\mathbb{R}^n)$ such that $\|f - f_\varepsilon\|_{L^1} < \varepsilon$ and $\|\partial_j f - \partial_j f_\varepsilon\|_{L^1} < \varepsilon$. Integrating

by parts gives

$$\begin{aligned}\widehat{\partial_j f_\varepsilon}(\xi) &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} \partial_j f_\varepsilon(x) e^{-i\xi \cdot x} dx \\ &= -(2\pi)^{-n/2} \int_{\mathbb{R}^n} f_\varepsilon(x) \partial_j (e^{-i\xi \cdot x}) dx \\ &= i\xi_j (2\pi)^{-n/2} \int_{\mathbb{R}^n} f_\varepsilon(x) e^{-i\xi \cdot x} dx\end{aligned}$$

so that $\widehat{\partial_j f_\varepsilon}(\xi) = i\xi_j \hat{f}_\varepsilon(\xi)$. Now we calculate:

$$\begin{aligned}|\widehat{\partial_j f}(\xi) - i\xi_j \hat{f}(\xi)| &= |\widehat{\partial_j f}(\xi) - \widehat{\partial_j f_\varepsilon}(\xi) + i\xi_j \hat{f}(\xi) - i\xi_j \hat{f}_\varepsilon(\xi)| \\ &\leq \|\partial_j f - \partial_j f_\varepsilon\|_{L^1} + |\xi| \|f - f_\varepsilon\|_{L^1} \\ &\leq \varepsilon(1 + |\xi|)\end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, we must have that $|\widehat{\partial_j f}(\xi) - i\xi_j \hat{f}(\xi)| = 0$, and the result follows.

- (2) Clearly we have that $x_j f \in L^1(\mathbb{R}^n)$, so $-ix_j \widehat{f}(\xi)$ is continuous. It suffices to prove then that

$$\Delta_j^{h_k} \hat{f}(\xi) \rightarrow -ix_j \widehat{f}(\xi), \text{ as } k \rightarrow \infty$$

for any sequence $\{h_k\}_{k \in \mathbb{N}} \subset \mathbb{R}$ with $h_k \rightarrow 0$. We calculate:

$$\begin{aligned}\Delta_j^{h_k} \hat{f}(\xi) &= \frac{1}{h_k} (\hat{f}(\xi + h_k e_j) - \hat{f}(\xi)) \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} \left(\frac{e^{-ix_j h_k} - 1}{h_k} \right) dx\end{aligned}$$

Now for $x \in \mathbb{R}^n$ we have:

$$f(x) e^{-ix \cdot \xi} \left(\frac{e^{-ix_j h_k} - 1}{h_k} \right) \rightarrow -ix_j f(x) e^{-ix \cdot \xi}$$

as $k \rightarrow \infty$. Noting that $|e^{i\theta} - 1| = 2|\sin \frac{\theta}{2}| \leq \theta$ for any $\theta \in \mathbb{R}$, we have that:

$$\left| f(x) e^{-ix \cdot \xi} \left(\frac{e^{-ix_j h_k} - 1}{h_k} \right) \right| \leq |x_j f(x)|$$

where the RHS is integrable. By the DCT, we have:

$$\lim_{k \rightarrow \infty} \Delta_j^{h_k} \hat{f}(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} -ix_j f(x) e^{-ix \cdot \xi} dx = -ix_j \widehat{f}(\xi).$$

We deduce that $\hat{f} \in C^1(\mathbb{R}^n)$.

□

Corollary 0.17.

- (1) Suppose $f \in C^k(\mathbb{R}^n)$ and $\partial^\alpha f \in L^1(\mathbb{R}^n)$ for $|\alpha| \leq k$. Then there is some constant $C_k > 0$ depending only on k such that:

$$\sup_{\xi \in \mathbb{R}^n} |(1 + |\xi|)^k \hat{f}(\xi)| \leq C_k \sum_{|\alpha| \leq k} \|\partial^\alpha f\|_{L^1}$$

- (2) Suppose $(1 + |x|)^k f \in L^1(\mathbb{R}^n)$. Then $\hat{f} \in C^k(\mathbb{R}^n)$ and for any $|\alpha| \leq k$ we have:

$$\sup_{\xi \in \mathbb{R}^n} |\partial^\alpha \hat{f}(\xi)| \leq (2\pi)^{-n/2} \|(1 + |x|)^k f\|_{L^1}$$

- (3) The Fourier transform is a continuous linear map from $\mathcal{S}$ into $\mathcal{S}$:

$$\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}$$

Proof.

- (1) First we note that for any k there is some constant C_k such that:

$$(1 + |\xi|)^k \leq C_k \sum_{|\alpha| \leq k} |\xi^\alpha f|$$

holds for any $\xi \in \mathbb{R}^n$. By Theorem 0.16 we know that:

$$i^{|\alpha|} \xi^\alpha \hat{f}(\xi) = \widehat{\partial^\alpha f}(\xi).$$

We therefore have:

$$(1 + |\xi|)^k |\hat{f}(\xi)| \leq C'_k \sum_{|\alpha| \leq k} |i^{|\alpha|} \xi^\alpha \hat{f}(\xi)| = C'_k \sum_{|\alpha| \leq k} |\widehat{\partial^\alpha f}(\xi)|$$

taking the supremum over $\xi \in \mathbb{R}^n$ and applying the Riemann-Lebesgue Lemma we conclude

$$\sup_{\xi \in \mathbb{R}^n} |(1 + |\xi|)^k \hat{f}(\xi)| \leq C_k \sum_{|\alpha| \leq k} \|\partial^\alpha f\|_{L^1}.$$

- (2) Again, by Theorem 0.16 we have that for $|\alpha| \leq k$

$$\partial^\alpha \hat{f}(\xi) = (-i)^{|\alpha|} \widehat{x^\alpha f}(\xi).$$

taking the supremum over $\xi \in \mathbb{R}^n$ and applying the Riemann-Lebesgue Lemma we conclude

$$\sup_{\xi \in \mathbb{R}^n} |\partial^\alpha \hat{f}(\xi)| \leq (2\pi)^{-n/2} \|x^\alpha f\|_{L^1} \leq (2\pi)^{-n/2} \|(1 + |x|)^k f\|_{L^1}$$

(3) Note that if

$$\sup_{x \in \mathbb{R}^n} (1 + |x|)^N |f(x)| < K,$$

we have:

$$\|f\|_{L^1} = \int_{\mathbb{R}^n} |f(x)| dx \leq K \int_{\mathbb{R}^n} \frac{1}{(1 + |x|)^N} dx < \infty$$

provided $N > n$. Thus in particular if $f \in \mathcal{S}$ then there exists some constant C_n such that:

$$\|(1 + |x|)^M \partial^\alpha f\|_{L^1} \leq C_n \sup_{x \in \mathbb{R}^n} (1 + |x|)^{M+n+1} |\partial^\alpha f(x)|$$

for all $M \in \mathbb{N}$ and all multi-indices α . Applying the previous two parts we conclude that $\hat{f} \in C^\infty(\mathbb{R}^n)$ and:

$$\sup_{\xi \in \mathbb{R}^n, |\beta| \leq M} (1 + |\xi|)^N |D^\beta \hat{f}(\xi)| \leq C_{N,M,n} \sup_{x \in \mathbb{R}^n, |\alpha| \leq N} (1 + |x|)^{M+n+1} |\partial^\alpha f(x)|$$

For some constant $C_{N,M,n}$ depending only on N, M, n . Thus $\hat{f} \in \mathcal{S}$. Moreover, if $\{f_j\}_{j \in \mathbb{N}} \subset \mathcal{S}$ is a sequence with $f_j \rightarrow 0$ in $\mathcal{S}$, then $\hat{f}_j \rightarrow 0$ in $\mathcal{S}$, so that $\mathcal{F}$ is continuous.

□

Proposition 0.18. Suppose $f, g \in L^1(\mathbb{R}^n)$, then

$$\int_{\mathbb{R}^n} \hat{f}(x) g(x) dx = \int_{\mathbb{R}^n} f(x) \hat{g}(x) dx$$

Proof. The integral is well-defined by the Riemann-Lebesgue Lemma. Moreover,

$$\begin{aligned} \int_{\mathbb{R}^n} \hat{f}(x) g(x) dx &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} f(y) e^{-ix \cdot y} dy \right) g(x) dx \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(y) \left(\int_{\mathbb{R}^n} g(x) e^{-ix \cdot y} dx \right) dy \\ &= \int_{\mathbb{R}^n} f(y) \hat{g}(y) dy \end{aligned}$$

□

Theorem 0.19 (Fourier inversion theorem). Suppose $f \in L^1(\mathbb{R}^n)$, and assume $\hat{f} \in L^1(\mathbb{R}^n)$, then for almost every x :

$$f(x) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} \hat{f}(\xi) e^{ix \cdot \xi} d\xi.$$

Proof. We shall establish the result by looking at the limit $\varepsilon \rightarrow 0$ of

$$I_\varepsilon(x) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} \hat{f}(\xi) e^{-\frac{1}{2}\varepsilon^2|\xi|^2} e^{ix \cdot \xi} d\xi.$$

in two different ways. First note that for $\xi \in \mathbb{R}^n$ we have

$$\hat{f}(\xi) e^{-\frac{1}{2}\varepsilon^2|\xi|^2} e^{ix \cdot \xi} \rightarrow \hat{f}(\xi) e^{ix \cdot \xi}$$

Moreover, we can estimate

$$|\hat{f}(\xi) e^{-\frac{1}{2}\varepsilon^2|\xi|^2} e^{ix \cdot \xi}| \leq |\hat{f}(\xi)|$$

Thus by the DCT we have:

$$I_\varepsilon(x) \rightarrow \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} \hat{f}(\xi) e^{ix \cdot \xi} d\xi, \text{ as } \varepsilon \rightarrow 0.$$

On the other hand, using Fubini's theorem:

$$\begin{aligned} I_\varepsilon(x) &= \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} f(y) e^{-i\xi \cdot y} dy \right) e^{-\frac{1}{2}\varepsilon^2|\xi|^2} e^{ix \cdot \xi} d\xi \\ &= \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} f(y) \left(\int_{\mathbb{R}^n} e^{-\frac{1}{2}\varepsilon^2|\xi|^2} e^{-i\xi \cdot (y-x)} d\xi \right) dy \\ &= \int_{\mathbb{R}^n} f(y) \frac{1}{\varepsilon^n (2\pi)^{\frac{n}{2}}} e^{-\frac{|y-x|^2}{2\varepsilon^2}} dy \\ &= f * \psi_\varepsilon(x) \end{aligned}$$

where $\psi_\varepsilon(x) = \varepsilon^{-n} \psi(\varepsilon^{-1}x)$ for

$$\psi(x) = \frac{1}{(2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2}|x|^2}.$$

Note that $\psi \in C^\infty(\mathbb{R}^n)$, $\psi(x) \geq 0$ and

$$\int_{\mathbb{R}^n} \psi(x) dx = 1$$

so by Theorem 0.9 we have that:

$$f * \psi_\varepsilon \rightarrow f,$$

in $L^1(\mathbb{R}^n)$, thus we must have that

$$f(x) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} \hat{f}(\xi) e^{ix \cdot \xi} d\xi$$

for almost every x . □

Theorem 0.20 (Plancherel Theorem). The mapping $f \rightarrow \hat{f}$ defined on $\mathcal{S}$ can be extended uniquely to a unitary operator on $L^2(\mathbb{R}^n)$.

Proof. Using the fact that $\mathcal{S}$ is dense in $L^2(\mathbb{R}^n)$ and the Fourier inversion theorem, it suffices to show that

$$\|\hat{\varphi}\|_{L^2} = \|\varphi\|_{L^2}, \quad \varphi \in \mathcal{S}$$

Let ψ be the function defined by

$$\psi(x) = \overline{\varphi(-x)}, \quad x \in \mathbb{R}^n.$$

Then $\psi \in \mathcal{S}$ and

$$\begin{aligned} \hat{\psi}(\xi) &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{-ix \cdot \xi} \overline{\varphi(-x)} dx \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \overline{\varphi(x)} dx = \overline{\hat{\varphi}(\xi)} \end{aligned}$$

Thus,

$$\begin{aligned} \|\varphi\|_{L^2}^2 &= \int_{\mathbb{R}^n} \varphi(x) \overline{\varphi(x)} dx = \int_{\mathbb{R}^n} \varphi(x) \psi(-x) dx \\ &= (\varphi * \psi)(0) \end{aligned}$$

Since $\varphi * \psi \in \mathcal{S}$. Hence we have

$$\begin{aligned} (\varphi * \psi)(0) &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} \widehat{\varphi * \psi}(\xi) d\xi = \int_{\mathbb{R}^n} \hat{\varphi}(\xi) \hat{\psi}(\xi) d\xi \\ &= \int_{\mathbb{R}^n} \hat{\varphi}(\xi) \overline{\hat{\varphi}(\xi)} d\xi = \|\hat{\varphi}\|_{L^2}^2 \end{aligned}$$

□

Definition 0.21. A sequence $\{\phi\}_{j \in \mathbb{N}}$ of rapidly decreasing functions tends to zero in $\mathcal{S}$ if for every multi-index α and $N \in \mathbb{N}$ we have:

$$\sup_{x \in \mathbb{R}^n} |(1 + |x|)^N \partial^\alpha \phi_j(x)| \rightarrow 0, \text{ as } j \rightarrow \infty$$

Similarly, $\phi_j \rightarrow \phi$ with respect to τ if $\phi_j - \phi \rightarrow 0$.

Definition 0.22. We define $\mathcal{S}'$ to be the continuous dual space of $\mathcal{S}$. That is to say, $u \in \mathcal{S}'$ if $u : \mathcal{S} \rightarrow \mathbb{C}$ is a continuous linear map. For a linear map $u : \mathcal{S} \rightarrow \mathbb{C}$ to belong to $\mathcal{S}'$, it is enough that:

$$\lim_{j \rightarrow \infty} u[\phi_j] \rightarrow 0,$$

for any sequence $\{\phi_j\}_{j \in \mathbb{N}} \subset \mathcal{S}$ which converges to zero in $\mathcal{S}$.

Definition 0.23. Suppose that $f \in L^1(\mathbb{R}^n)$, and $\phi \in \mathcal{S}$. Then since $\hat{f} \in C^0(\mathbb{R}^n)$ and $\hat{f}$ decays towards infinity, we have that $T_{\hat{f}} \in \mathcal{S}'$. By Fubini we have:

$$\begin{aligned} T_{\hat{f}}[\phi] &= \int_{\mathbb{R}^n} \hat{f}(x) \phi(x) dx = \int_{\mathbb{R}^n} (2\pi)^{-n/2} \left(\int_{\mathbb{R}^n} f(y) e^{-ix \cdot y} dy \right) \phi(x) dx \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(y) \left(\int_{\mathbb{R}^n} \phi(x) e^{-ix \cdot y} dx \right) dy \\ &= \int_{\mathbb{R}^n} f(x) \hat{\phi}(x) dx. \end{aligned}$$

Thus for $f \in L^1(\mathbb{R}^n)$ we have that:

$$T_{\hat{f}}[\phi] = T_f[\hat{\phi}], \text{ for all } \phi \in \mathcal{S}.$$

For a distribution $u \in \mathcal{S}'$, we define the Fourier transform of u , written $\hat{u} \in \mathcal{S}'$ to be the distribution satisfying:

$$\hat{u}[\phi] = u[\hat{\phi}], \text{ for all } \phi \in \mathcal{S}.$$

Examples.

(1) For $\xi \in \mathbb{R}^n$ we have:

$$\widehat{\delta_\xi} = (2\pi)^{-n/2} T_{e_{-\xi}}$$

To see this, we use the definition. For $\phi \in \mathcal{S}$:

$$\widehat{\delta_\xi}[\phi] = \delta_\xi[\hat{\phi}] = \hat{\phi}(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-ix \cdot \xi} \phi(x) dx = (2\pi)^{-n/2} T_{e_{-\xi}}[\phi]$$

Since ϕ was arbitrary, the distributions are equal.

(2) For $x \in \mathbb{R}^n$ we have:

$$\widehat{T_{e_x}}[\phi] = T_{e_x}[\hat{\phi}] = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \hat{\phi}(\xi) d\xi = \phi(x) = \delta_x[\phi].$$

Again, as ϕ is arbitrary the distributions are equal. Note that a particular case is $\widehat{T_1} = \delta_0$.

(3) For α a multi-index, denoted by X^α the map

$$X^\alpha : x \mapsto x^\alpha.$$

Then we have:

$$\widehat{T_{X^\alpha}} = i^{|\alpha|} \partial^\alpha \delta_0$$

For $\phi \in \mathcal{S}$:

$$\begin{aligned} \widehat{T_{X^\alpha}}[\phi] &= T_{X^\alpha}[\hat{\phi}] = (2\pi)^{-n/2} \int_{\mathbb{R}^n} \xi^\alpha \hat{\phi}(\xi) d\xi \\ &= (-i)^{|\alpha|} (2\pi)^{-n/2} \int_{\mathbb{R}^n} \widehat{\partial^\alpha \phi}(\xi) d\xi \\ &= (-i)^{|\alpha|} \partial^\alpha \phi(0) = i^{|\alpha|} \times (-1)^{|\alpha|} \delta_0[\partial^\alpha \phi] \\ &= i^{|\alpha|} \partial^\alpha \delta_0[\phi] \end{aligned}$$

Definition 0.24. Recall that if $\phi \in \mathcal{S}$, then $\tau_x \phi \in \mathcal{S}$ is the translate of ϕ , given by $\tau_x \phi(y) = \phi(y - x)$, and $\check{\phi} \in \mathcal{S}$ is given by $\check{\phi}(y) = \phi(-y)$. For $u \in \mathcal{S}$, we define:

$$\tau_x u[\phi] = u[\tau_{-x} \phi], \quad \check{u}[\phi] = u[\check{\phi}]$$

Notice also that if $f \in C^\infty(\mathbb{R}^n)$ is a function of tempered growth, i.e., for every α and there exists a constant C_α and integer N_α such that:

$$|\partial^\alpha f(x)| \leq C_\alpha (1 + |x|)^{N_\alpha}, \quad \forall x \in \mathbb{R}^n.$$

then $\phi f \in \mathcal{S}$ when $\phi \in \mathcal{S}$ and we can define $fu \in \mathcal{S}'$ by

$$fu[\phi] = u[f\phi]$$

Lemma 0.25. Suppose $u \in \mathcal{S}'$ is a tempered distribution. Then:

$$\widehat{e_x u} = \tau_x \hat{u}, \quad \widehat{\tau_x u} = e_{-x} \hat{u}$$

and

$$\widehat{\partial^\alpha u} = i^{|\alpha|} X^\alpha \hat{u}, \quad \partial^\alpha \hat{u} = (-i)^{|\alpha|} \widehat{X^\alpha u}$$

Moreover:

$$\hat{\check{u}} = \check{\hat{u}}$$

so that the Fourier transform on $\mathcal{S}'$ is invertible.

Proof. These are all calculation using the corresponding results for $\mathcal{S}$. Take $\phi \in \mathcal{S}$, we have:

$$\widehat{e_x u}[\phi] = e_x u[\hat{\phi}] = u[e_x \hat{\phi}] = u[\widehat{\tau_{-x} \phi}] = \hat{u}[\tau_{-x} \phi] = \tau_x u[\phi].$$

Since ϕ was arbitrary, we have $\widehat{e_x u} = \tau_x \hat{u}$. Similarly, we calculate:

$$\widehat{\tau_x u}[\phi] = \tau_x u[\hat{\phi}] = u[\tau_{-x} \hat{\phi}] = u[\widehat{e_{-x} \phi}] = \hat{u}[e_{-x} \phi] = e_{-x} u[\phi].$$

Next we have

$$\begin{aligned} \widehat{\partial^\alpha u}[\phi] &= \partial^\alpha u[\hat{\phi}] = (-1)^{|\alpha|} u[\partial^\alpha \hat{\phi}] \\ &= (-1)^{|\alpha|} u[(-i)^{|\alpha|} \widehat{X^\alpha \phi}] = i^{|\alpha|} u[\widehat{X^\alpha \phi}] \\ &= i^{|\alpha|} \hat{u}[X^\alpha \phi] = (i^{|\alpha|} X^\alpha \hat{u})[\phi] \end{aligned}$$

Similarly:

$$\begin{aligned} \partial^\alpha \hat{u}[\phi] &= (-1)^{|\alpha|} \hat{u}[\partial^\alpha \phi] = (-1)^{|\alpha|} u[\widehat{\partial^\alpha \phi}] \\ &= (-1)^{|\alpha|} u[i^{|\alpha|} X^\alpha \hat{\phi}] = (-i)^{|\alpha|} X^\alpha u[\hat{\phi}] \\ &= ((-i)^{|\alpha|} \widehat{X^\alpha u})[\phi] \end{aligned}$$

Finally we have

$$\hat{\hat{u}}(\phi) = \hat{u}(\hat{\phi}) = u[\hat{\hat{\phi}}] = u[\check{\phi}] = (\check{u}[\phi])$$

Since $\check{\check{u}} = u$, we have that the Fourier transform is invertible. $\square$

Lemma 0.26. The map

$$\begin{aligned}\mathcal{F} : \mathcal{S}' &\rightarrow \mathcal{S}' \\ u &\mapsto \hat{u}\end{aligned}$$

is a linear homeomorphism.

Proof. We have seen that $\mathcal{F}$ is linear. From the definition of the weak-* topology, a sequence $\{u_j\}_{j \in \mathbb{N}} \subset \mathcal{S}'$ converges to u if

$$u_j[\phi] \rightarrow u[\phi]$$

for all $\phi \in \mathcal{S}$. Suppose that we have such a convergent sequence in $\mathcal{S}'$. We calculate:

$$\hat{u}_j[\phi] = u_j[\hat{\phi}] \rightarrow u[\hat{\phi}] = \hat{u}[\phi].$$

Thus if $u_j \rightarrow u$ we have $\mathcal{F}(u_j) \rightarrow \mathcal{F}(u)$. Thus $\mathcal{F}$ is continuous. Since $\mathcal{F}^4 = \text{id}$, we have that $\mathcal{F}$ is invertible and the inverse is also continuous. $\square$

Lemma 0.27. For any two multi-indices α and β ,

$$\partial^\beta x^\alpha = \begin{cases} \beta! \binom{\alpha}{\beta} x^{\alpha-\beta}, & \beta \leq \alpha \\ 0, & \text{otherwise.} \end{cases}$$

Proof. Fix α and use induction on $|\beta|$. The base case $\beta = 0$ clearly holds. Assume $\beta > 0$ and the equation holds for all multi-indices of size less than β . $\beta > 0$ implies $\beta_i \geq 1$ for some $i \in \{1, \dots, n\}$. Let δ be the multi-index with a 1 at position i and 0 at all other positions. Then $\delta \leq \beta$, so we can let $\gamma = \beta - \delta$, and then we see that $\beta = \gamma + \delta$ and $\gamma < \beta$. The induction assumption now holds for γ . So now we want to show that it also holds for β . There are three cases:

- (i) Suppose that $\gamma \not\leq \alpha$ and $\beta \not\leq \alpha$. Then $\partial^\gamma x^\alpha = 0$ by the induction assumption, so

$$\partial^\beta x^\alpha = \partial^{\gamma+\delta} x^\alpha = \partial_i \partial^\gamma x^\alpha = \partial_i 0 = 0,$$

which shows that the equation holds for β .

- (ii) Suppose that $\gamma \leq \alpha$ and $\beta \not\leq \alpha$. The latter implies that $\beta_k > \alpha_k$ for some k . But $\beta = \gamma + \delta$, so $\beta_j = \gamma_j \leq \alpha_j$ for all $j \neq i$. Therefore, the only possible option is that $k = i$. Since $\beta_i = \gamma_i + 1$, we see that

$$\gamma_i \leq \alpha_i < \beta_i = \gamma_i + 1 \implies \gamma_i = \alpha_i.$$

By the induction assumption,

$$\partial^\gamma x^\alpha = \gamma! \binom{\alpha}{\gamma} x^{\alpha-\gamma}.$$

$\alpha_i - \gamma_i$ so $x^{\alpha-\gamma}$ is independent of x_i . Since $\partial^\beta = \partial_i \partial^\gamma$, we then have that $\partial^\beta x^\alpha = 0$.

- (iii) Suppose that $\gamma \leq \alpha$ and $\beta \leq \alpha$. Since $\beta_i = \gamma_i + 1$, we then see that $\alpha_i - \gamma_i > 0$. Therefore,

$$\begin{aligned}
\partial_i(x^{\alpha-\gamma}) &= \partial_i(x^{\alpha_1-\gamma_1} \dots x^{\alpha_i-\gamma_i} \dots x^{\alpha_n-\gamma_n}) \\
&= (\alpha_i - \gamma_i)x^{\alpha_1-\gamma_1} \dots x^{\alpha_i-\gamma_i-1} \dots x^{\alpha_n-\gamma_n} \\
&= (\alpha_i - \gamma_i)x^{\alpha_1-\gamma_1} \dots x^{\alpha_i-\beta_i} \dots x^{\alpha_n-\gamma_n} \\
&= (\alpha_i - \gamma_i)x^{\alpha-\beta}.
\end{aligned}$$

Then by the induction assumption

$$\partial^\beta x^\alpha = \partial_i \partial^\gamma x^\alpha = \gamma! \binom{\alpha}{\gamma} \partial_i(x^{\alpha-\gamma}) = \gamma! \binom{\alpha}{\gamma} (\alpha_i - \gamma_i) x^{\alpha-\beta}.$$

Notice that

$$\begin{aligned}
\gamma_i! \binom{\alpha_i}{\gamma_i} (\alpha_i - \gamma_i) &= \frac{\alpha_i!}{(\alpha_i - \gamma_i)!} (\alpha_i - \gamma_i) \\
&= \frac{\alpha_i!}{(\alpha_i - \gamma_i - 1)!} \\
&= \frac{\alpha_i!}{(\alpha_i - \beta_i)!} \\
&= \beta_i! \binom{\alpha_i}{\beta_i}.
\end{aligned}$$

We then obtain

$$\begin{aligned}
\gamma! \binom{\alpha}{\gamma} (\alpha_i - \gamma_i) &= \gamma_i! \binom{\alpha_i}{\gamma_i} (\alpha_i - \gamma_i) \prod_{\substack{j=1 \\ j \neq i}}^n \gamma_j! \binom{\alpha_j}{\gamma_j} \\
&= \beta_i! \binom{\alpha_i}{\beta_i} \prod_{\substack{j=1 \\ j \neq i}}^n \beta_j! \binom{\alpha_j}{\beta_j} \\
&= \beta! \binom{\alpha}{\beta}.
\end{aligned}$$

Thus, we see that

$$\partial^\beta x^\alpha = \beta! \binom{\alpha}{\beta} x^{\alpha-\beta},$$

which completes the induction step.

□

Lemma 0.28. For any $N \in \mathbb{N}$,

$$|z|^{2N} \leq n^N \sum_{|\gamma|=N} |z^\gamma|^2, \quad z \in \mathbb{R}^n.$$

Proof. We use induction on N . For $N = 0$, we have $|z|^{2N} = 1$ and $n^N \sum_{|\gamma|=N} |z^\gamma|^2 = 1$ so the base case holds. Now suppose that $N > 0$ and that the inequality holds for $N - 1$. Then we have

$$|z|^{2N} = |z|^2 |z|^{2(N-1)} \leq |z|^2 n^{N-1} \sum_{|\gamma|=N-1} |z^\gamma|^2.$$

Notice that there are exactly n different multi-indices δ with $|\delta| = 1$. Also, $z^\delta = z_i$ where i is the index of the only non-zero component of δ . Hence,

$$|z|^2 = \sum_{i=1}^n z_i^2 = \sum_{|\delta|=1} |z^\delta|^2.$$

Thus,

$$\begin{aligned} |z|^{2N} &\leq n^{N-1} \sum_{|\delta|=1} |z^\delta|^2 \sum_{|\gamma|=N-1} |z^\gamma|^2 = n^{N-1} \sum_{|\gamma|=N-1} \sum_{|\delta|=1} |z^\gamma|^2 |z^\delta|^2 \\ &= n^{N-1} \sum_{|\gamma|=N-1} \sum_{|\delta|=1} |z^{\gamma+\delta}|^2. \end{aligned}$$

This is equivalent to

$$n^{N-1} \sum_{|\gamma|=N-1} \sum_{|\delta|=1} |z^{\gamma+\delta}|^2 = \sum_{|\alpha|=N} \sum_{\alpha_i \geq 1} |z^\alpha|^2.$$

For each multi-index α in the sum, the number of terms of the form $|z^\alpha|^2$ is just the number of indices $i \in \{1, \dots, n\}$ that satisfy $\alpha_i \geq 1$, which is less than or equal to n . Thus,

$$|z|^{2N} \leq n^{N-1} \sum_{|\alpha|=N} \sum_{\alpha_i \geq 1} |z^\alpha|^2 \leq n^{N-1} \sum_{|\alpha|=N} n |z^\alpha|^2 = n^N \sum_{|\alpha|=N} |z^\alpha|^2$$

which completes the induction step. $\square$

1 Introduction to Pseudo-Differential Operators

1.1 Symbols

Definition 1.1. Let $m \in (-\infty, \infty)$. Then we define S^m to be the set of all functions $\sigma(x, \xi)$ in $C^\infty(\mathbb{R}^n \times \mathbb{R}^n)$ such that for any two multi-indices α and β , there is positive constant $C_{\alpha, \beta}$, depending on α and β only, such that

$$\left| D_x^\alpha D_\xi^\beta \sigma(x, \xi) \right| \leq C_{\alpha, \beta} (1 + |\xi|)^{m - |\beta|}, \quad x, \xi \in \mathbb{R}^n.$$

We call any function σ in $\bigcup_{m \in \mathbb{R}} S^m$ a symbol.

Proposition 1.2. Symbols satisfy the following properties:

- (1) S^m is a vector space.
- (2) If $\sigma \in S^{m_1}$ and $\tau \in S^{m_2}$, then $\sigma\tau \in S^{m_1+m_2}$.
- (3) If $\sigma \in S^m$, then for all multi-indices α and β , $\partial_x^\alpha \partial_\xi^\beta \sigma \in S^{m-|\beta|}$.
- (4) If $m_1 \leq m_2$, then $S^{m_1} \subseteq S^{m_2}$.
- (5) If $\sigma \in S^m$, $\varphi \in \mathcal{S}$, and $\tau(x, \xi) = \sigma(x, \xi)\varphi(\xi)$, then $\tau \in \bigcap_{k \in \mathbb{R}} S^k$.
- (6) If $\varphi \in \mathcal{S}$ and $\tau(x, \xi) = \varphi(\xi)$, then $\tau \in \bigcap_{m \in \mathbb{R}} S^m$.

Proof.

- (1) Clearly $0 \in S^m$. If $\sigma, \tau \in S^m$, then for all multi-indices α and β there are constants $A_{\alpha, \beta}, B_{\alpha, \beta}$ such that

$$\begin{aligned} |D_x^\alpha D_\xi^\beta \sigma(x, \xi)| &\leq A_{\alpha, \beta} (1 + |\xi|)^{m - |\beta|} \\ |D_x^\alpha D_\xi^\beta \tau(x, \xi)| &\leq B_{\alpha, \beta} (1 + |\xi|)^{m - |\beta|} \end{aligned}$$

By the triangle inequality,

$$\begin{aligned} |D_x^\alpha D_\xi^\beta (\sigma + \tau)(x, \xi)| &\leq A_{\alpha, \beta} (1 + |\xi|)^{m - |\beta|} + B_{\alpha, \beta} (1 + |\xi|)^{m - |\beta|} \\ &\leq (A_{\alpha, \beta} + B_{\alpha, \beta}) (1 + |\xi|)^{m - |\beta|}. \end{aligned}$$

So $\sigma + \tau \in S^m$. If $\lambda \in \mathbb{C}$, then

$$|D_x^\alpha D_\xi^\beta (\lambda\sigma)(x, \xi)| = |\lambda| |D_x^\alpha D_\xi^\beta \sigma(x, \xi)| \leq |\lambda| A_{\alpha, \beta} (1 + |\xi|)^{m - |\beta|}.$$

So $\lambda\sigma \in S^m$. Thus S^m is a vector space.

(2) For all multi-indices α and β , by Leibnitz's formula: $|D_x^\alpha D_\xi^\beta(\sigma\tau)(x, \xi)|$

$$\begin{aligned} &= \left| D_x^\alpha \left(\sum_{\delta \leq \beta} \binom{\beta}{\delta} D_\xi^\delta \sigma(x, \xi) D_\xi^{\beta-\delta} \tau(x, \xi) \right) \right| \\ &= \left| \sum_{\delta \leq \beta} \sum_{\gamma \leq \alpha} \binom{\beta}{\delta} \binom{\alpha}{\gamma} D_x^\gamma D_\xi^\delta \sigma(x, \xi) D_x^{\alpha-\gamma} D_\xi^{\beta-\delta} \tau(x, \xi) \right| \\ &\leq \sum_{\delta \leq \beta} \sum_{\gamma \leq \alpha} \binom{\beta}{\delta} \binom{\alpha}{\gamma} |D_x^\gamma D_\xi^\delta \sigma(x, \xi)| |D_x^{\alpha-\gamma} D_\xi^{\beta-\delta} \tau(x, \xi)|. \end{aligned}$$

By assumption, there are constants $A_{\gamma, \delta}$ and $B_{\alpha, \beta, \gamma, \delta}$ such that

$$\begin{aligned} |D_x^\gamma D_\xi^\delta \sigma(x, \xi)| &\leq A_{\gamma, \delta} (1 + |\xi|)^{m_1 - |\delta|} \\ |D_x^{\alpha-\gamma} D_\xi^{\beta-\delta} \tau(x, \xi)| &\leq B_{\alpha, \beta, \gamma, \delta} (1 + |\xi|)^{m_2 - |\beta| + |\delta|}. \end{aligned}$$

Thus,

$$\begin{aligned} |D_x^\alpha D_\xi^\beta(\sigma\tau)(x, \xi)| &\leq \sum_{\delta \leq \beta} \sum_{\gamma \leq \alpha} \binom{\beta}{\delta} \binom{\alpha}{\gamma} A_{\gamma, \delta} B_{\alpha, \beta, \gamma, \delta} (1 + |\xi|)^{m_1 + m_2 - |\beta|} \\ &\leq C_{\alpha, \beta} (1 + |\xi|)^{m_1 + m_2 - |\beta|} \end{aligned}$$

where

$$C_{\alpha, \beta} = \sum_{\delta \leq \beta} \sum_{\gamma \leq \alpha} \binom{\beta}{\delta} \binom{\alpha}{\gamma} A_{\gamma, \delta} B_{\alpha, \beta, \gamma, \delta}.$$

Hence, $\sigma\tau \in S^{m_1 + m_2}$.

(3) Fix α and β . Then by definition of S^m ,

$$\begin{aligned} |D_x^\gamma D_\xi^\delta (D_x^\alpha D_\xi^\beta \sigma)(x, \xi)| &= |D_x^{\alpha+\gamma} D_\xi^{\beta+\delta} \sigma(x, \xi)| \\ &\leq C_{\alpha+\gamma, \beta+\delta} (1 + |\xi|)^{m - |\beta| - |\delta|}. \end{aligned}$$

Since α and β are fixed, $C_{\alpha+\gamma, \beta+\delta}$ depends only on γ and δ , so $D_x^\alpha D_\xi^\beta \sigma \in S^{m-|\beta|}$ by definition.

(4) This follows immediately by definition and the fact that $m_1 \leq m_2$ implies

$$(1 + |\xi|)^{m_1 - |\beta|} \leq (1 + |\xi|)^{m_2 - |\beta|}.$$

(5) Let $k \in \mathbb{R}$ be arbitrary. By Leibniz's formula, for all multi-indices α and β ,

$$|D_x^\alpha D_\xi^\beta \tau(x, \xi)| \leq \sum_{\gamma \leq \beta} \binom{\beta}{\gamma} |D_x^\alpha D_\xi^\gamma \sigma(x, \xi)| |D_\xi^{\beta-\gamma} \varphi(\xi)|.$$

Since $\sigma \in S^m$, there is a constant $A_{\alpha,\gamma}$ such that

$$|D_x^\alpha D_\xi^\gamma \sigma(x, \xi)| \leq A_{\alpha,\gamma} (1 + |\xi|)^{m-|\gamma|}.$$

Since $\varphi \in \mathcal{S}$, there is a constant $B_{\beta,\gamma}$ such that

$$(1 + |\xi|)^{m+|\beta|-|\gamma|-k} |D^{\beta-\gamma} \varphi(\xi)| \leq B_{\beta,\gamma}.$$

Therefore,

$$\begin{aligned} |D_x^\alpha D_\xi^\beta \tau(x, \xi)| &\leq \sum_{\gamma \leq \beta} \binom{\beta}{\gamma} A_{\alpha,\gamma} (1 + |\xi|)^{m-|\gamma|} B_{\beta,\gamma} (1 + |\xi|)^{k+|\gamma|-|\beta|-m} \\ &= C_{\alpha,\beta} (1 + |\xi|)^{k-|\beta|} \end{aligned}$$

where

$$C_{\alpha,\beta} = \sum_{\gamma \leq \beta} \binom{\beta}{\gamma} A_{\alpha,\gamma} B_{\beta,\gamma}$$

i.e. $\tau \in S^k$. Since k was arbitrary, this implies that $\tau \in \bigcap_{k \in \mathbb{R}} S^k$.

(6) This is a special case of (5) for $\sigma = 1$.

□

Definition 1.3. Let σ be a symbol. The pseudo-differential operator T_σ associated to σ is defined by

$$(T_\sigma \varphi)(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \sigma(x, \xi) \hat{\varphi}(\xi) d\xi, \quad \varphi \in \mathcal{S}.$$

Example. Let $P(x, D) = \sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha$ be a linear partial differential operator on $\mathbb{R}^n$. If all the coefficients $a_\alpha(x)$ are C^∞ and have bounded derivatives of all orders, then the polynomial $P(x, \xi) = \sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha$ is in S^m and hence $P(x, D)$ is a pseudo-differential operator.

Proof. First note that

$$\begin{aligned} (P(x, D)\varphi)(x) &= \sum_{|\alpha| \leq m} a_\alpha(x) (D^\alpha \varphi)(x) = \sum_{|\alpha| \leq m} a_\alpha(x) (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \widehat{D^\alpha \varphi} d\xi \\ &= \sum_{|\alpha| \leq m} a_\alpha(x) (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \xi^\alpha \hat{\varphi} d\xi \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \left(\sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha \right) \hat{\varphi}(\xi) d\xi \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} P(x, \xi) \hat{\varphi}(\xi) d\xi \end{aligned}$$

Let γ and δ be multi-indices. Then

$$|(D_x^\gamma D_\xi^\delta P)| \leq \sum_{|\alpha| \leq m} C_{\alpha, \gamma} |\partial_\xi^\delta \xi^\alpha|$$

for all $x, \xi \in \mathbb{R}^n$, where $C_{\alpha, \gamma} = \sup_{x \in \mathbb{R}^n} |(D_x^\gamma a_\alpha)(x)|$. By Lemma 0.27, we have

$$\begin{aligned} |(D_x^\gamma D_\xi^\delta P)(x, \xi)| &\leq \sum_{|\alpha| \leq m} C_{\alpha, \gamma} \delta! \binom{\alpha}{\delta} |\xi|^{|\alpha| - |\delta|} \\ &\leq C_{\gamma, \delta} (1 + |\xi|)^{m - |\delta|} \end{aligned}$$

for all $x, \xi \in \mathbb{R}^n$, where $C_{\gamma, \delta} = \sum_{|\alpha| \leq m} C_{\alpha, \gamma} \delta! \binom{\alpha}{\delta}$. $\square$

Lemma 1.4. Let f be a continuous tempered function such that

$$T_f(\varphi) = 0, \quad \varphi \in \mathcal{S}.$$

Then f is identically zero on $\mathbb{R}^n$.

Proof. Without loss of generality, we assume that f is real-valued. Suppose $f(x_0) \neq 0$ for some $x_0 \in \mathbb{R}^n$. Then there is an open ball $B(x_0, r)$ with centred x_0 and radius r on which f is strictly positive (or negative). Choose a non-zero function $\varphi_0 \in C_0^\infty(\mathbb{R}^n)$ such that $\varphi_0(x) \geq 0$ for all $x \in \mathbb{R}^n$ and $\text{supp}(\varphi_0) \subseteq B(x_0, r)$ such function exists by Theorem 0.1. It is clear that $\varphi_0 \in \mathcal{S}$ and

$$T_f(\varphi_0) = \int_{\mathbb{R}^n} f(x) \varphi_0(x) dx$$

is strictly positive (or negative) which contradicts $T_f(\varphi) = 0$. $\square$

Proposition 1.5. Let σ and τ be two symbols such that $T_\sigma = T_\tau$. Then $\sigma = \tau$.

Proof. We have

$$\int_{\mathbb{R}^n} e^{ix \cdot \xi} (\sigma(x, \xi) - \tau(x, \xi)) \hat{\varphi}(\xi) d\xi = 0, \quad \xi \in \mathcal{S}$$

Since $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}$ is bijective, it follows that

$$\int_{\mathbb{R}^n} e^{ix \cdot \xi} (\sigma(x, \xi) - \tau(x, \xi)) \varphi(\xi) d\xi = 0, \quad \xi \in \mathcal{S}$$

Now, for any fixed $x \in \mathbb{R}^n$, $e^{ix \cdot \xi} (\sigma(x, \xi) - \tau(x, \xi))$ is a continuous tempered function in the variable ξ . Hence, by the Lemma above, $e^{ix \cdot \xi} (\sigma(x, \xi) - \tau(x, \xi)) = 0$ for all $\xi \in \mathbb{R}^n$. Since x is fixed but arbitrary, it follows that $\sigma(x, \xi) - \tau(x, \xi) = 0$ for all $x, \xi \in \mathbb{R}^n$ i.e. $\sigma = \tau$. $\square$

Proposition 1.6. Let σ be a symbol. Then T_σ maps the Schwartz space $\mathcal{S}$ into self.

Proof. Let $\varphi \in \mathcal{S}$. Then for any two multi-indices α and β , we need only prove that

$$\sup_{x \in \mathbb{R}^n} |x^\alpha (D^\beta (T_\sigma \varphi))(x)| < \infty.$$

But, using integration by parts and Leibnitz's formula, we have $x^\alpha (D^\alpha (T_\sigma \varphi))(x)$

$$\begin{aligned} &= x^\alpha (2\pi)^{-n/2} \int_{\mathbb{R}^n} D_x^\beta (e^{ix \cdot \xi} \sigma(x, \xi)) \hat{\varphi}(\xi) d\xi \\ &= x^\alpha (2\pi)^{-n/2} \int_{\mathbb{R}^n} \sum_{\gamma \leq \beta} \binom{\beta}{\gamma} \xi^\gamma e^{ix \cdot \xi} (D_x^{\beta-\gamma} \sigma)(x, \xi) \hat{\varphi}(\xi) d\xi \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} \sum_{\gamma \leq \beta} \binom{\beta}{\gamma} \xi^\gamma (D_\xi^\alpha e^{ix \cdot \xi}) (D_x^{\beta-\gamma} \sigma)(x, \xi) \hat{\varphi}(\xi) d\xi \\ &= (2\pi)^{-n/2} (-1)^{|\alpha|} \int_{\mathbb{R}^n} \sum_{\gamma \leq \beta} \binom{\beta}{\gamma} e^{ix \cdot \xi} D_\xi^\alpha \{ (D_x^{\beta-\gamma} \sigma)(x, \xi) \xi^\gamma \hat{\varphi}(\xi) \} d\xi \\ &= (2\pi)^{-n/2} (-1)^{|\alpha|} \int_{\mathbb{R}^n} \sum_{\gamma \leq \beta} \sum_{\delta \leq \alpha} \binom{\beta}{\gamma} \binom{\alpha}{\delta} e^{ix \cdot \xi} (D_\xi^{\alpha-\delta} D_x^{\beta-\gamma} \sigma)(x, \xi) D_\xi^\delta (\xi^\gamma \hat{\varphi}(\xi)) d\xi. \end{aligned}$$

Using the fact that σ is a symbol, say $\sigma \in S^m$, we can find positive constants $C_{\alpha, \beta, \gamma, \delta}$, depending on α, β, γ and δ only, such that $\sup_{x \in \mathbb{R}^n} |x^\alpha (D^\beta (T_\sigma \varphi))(x)|$

$$\leq (2\pi)^{-n/2} \sum_{\gamma \leq \beta} \sum_{\delta \leq \alpha} \binom{\beta}{\gamma} \binom{\alpha}{\delta} C_{\alpha, \beta, \gamma, \delta} \int_{\mathbb{R}^n} (1 + |\xi|)^{m-|\alpha|+|\delta|} |D_\xi^\delta (\xi^\gamma \hat{\varphi}(\xi))| d\xi.$$

Since $\varphi \in \mathcal{S}$, it follows that

$$\sup_{x \in \mathbb{R}^n} |x^\alpha (D^\beta (T_\sigma \varphi))(x)| < \infty,$$

and hence the proof is complete. $\square$

Definition 1.7. Let $\sigma \in S^m$. Suppose we can find $\sigma_j \in S^{m_j}$ where

$$m = m_0 > m_1 > m_2 > \cdots > m_j \rightarrow -\infty, \quad j \rightarrow \infty$$

such that

$$\sigma - \sum_{j=0}^{N-1} \sigma_j \in S^{m_N}$$

for every positive integer N . Then we call $\sum_{j=0}^{\infty} \sigma_j$ as asymptotic expansion of σ and we write

$$\sigma \sim \sum_{j=0}^{\infty} \sigma_j.$$

Proposition 1.8. There exists a function $\psi \in C^\infty(\mathbb{R}^n)$ such that $0 \leq \psi(\xi) \leq 1$ for $\xi \in \mathbb{R}^n$, $\psi(\xi) = 0$ for $|\xi| \leq 1$ and $\psi(\xi) = 1$ for $|\xi| \geq 2$.

Proof. We need to construct a function $\varphi_0 \in C_0^\infty(\mathbb{R}^n)$ such that $0 \leq \varphi_0(\xi) \leq 1$ for $\xi \in \mathbb{R}^n$, $\varphi_0(\xi) = 1$ for $|\xi| \leq 1$ and $\varphi_0(\xi) = 0$ for $|\xi| \geq 2$. For then the function $\psi = 1 - \varphi_0$ will satisfy all the conditions required. To construct φ_0 , let f be any continuous function on $\mathbb{R}^n$ such that $0 \leq f(t) \leq 1$ for $t \in \mathbb{R}^n$, $f(t) = 1$ for $|t| \leq \frac{3}{2}$ and $f(t) = 0$ for $|t| \geq \frac{7}{4}$. Let $\varphi \in C_0^\infty(\mathbb{R}^n)$ be a non-negative function such that $\varphi(s) = 0$ for $|s| \geq \frac{1}{4}$ and

$$\int_{|s| \leq \frac{1}{4}} \varphi(s) ds = 1.$$

such function exists by Theorem 0.1. Let $\varphi_0 = f * \varphi$ then $\varphi_0 \in C^\infty(\mathbb{R}^n)$ by Theorem 0.9. Furthermore, we see that $\varphi_0(t) = 0$ for $|t| \geq 2$ by Lemma 0.8. Finally, for $|t| \leq 1$,

$$\varphi_0(t) = \int_{\mathbb{R}^n} f(t-s)\varphi(s)ds = \int_{|s| \leq \frac{1}{4}} f(t-s)\varphi(s)ds.$$

Since, for $|t| \leq 1$ and $|s| \leq \frac{1}{4}$, we have $|s-t| \leq \frac{5}{4}$ and hence $f(t-s) = 1$. Therefore, $\varphi_0(t) = 1$ for $|t| \leq 1$. It's clear that $0 \leq \varphi_0(t) \leq 1$ for all $t \in \mathbb{R}^n$ and we are done. $\square$

Theorem 1.9. Let $m_0 > m_1 > m_2 > \dots > m_j \rightarrow -\infty$ as $j \rightarrow \infty$. Suppose $\sigma_j \in S^{m_j}$. Then there exists symbol $\sigma \in S^{m_0}$ such that $\sigma \sim \sum_{j=0}^\infty \sigma_j$. Moreover, if τ is another symbol with the same asymptotic expansion, then $\sigma - \tau \in \bigcap_{m \in \mathbb{R}} S^m$.

Proof. let $\psi \in C^\infty(\mathbb{R}^n)$ be such that $0 \leq \psi(\xi) \leq 1$ for $\xi \in \mathbb{R}^n$, $\psi(\xi) = 0$ for $|\xi| \leq 1$ and $\psi(\xi) = 1$ for $|\xi| \geq 2$ which exists by Proposition 1.8. Let $\{\varepsilon_j\}$ be a sequence of positive numbers such that $1 > \varepsilon_0 > \varepsilon_1 > \dots > \varepsilon_j \rightarrow 0$ as $j \rightarrow \infty$. Define the function σ on $\mathbb{R}^n \times \mathbb{R}^n$ by

$$\sigma(x, \xi) := \sum_{j=0}^\infty \psi(\varepsilon_j \xi) \sigma_j(x, \xi), \quad x, \xi \in \mathbb{R}^n.$$

Note that for each $(x_0, \xi_0) \in \mathbb{R}^n \times \mathbb{R}^n$, there exists a neighbourhood U of (x_0, ξ_0) and a positive integer N such that $\psi(\varepsilon_j \xi) \sigma_j(x, \xi) = 0$ for all $(x, \xi) \in U$ and $j > N$. Hence $\sigma \in C^\infty(\mathbb{R}^n \times \mathbb{R}^n)$. Furthermore, for any $\varepsilon \in (0, 1]$ and non-zero multi-index α , $\psi(\varepsilon \xi) = 0$ if $|\xi| \leq \frac{1}{\varepsilon}$, $\psi(\varepsilon \xi) = 1$ if $|\xi| \geq \frac{2}{\varepsilon}$; $\partial_\xi^\alpha(\psi(\varepsilon \xi)) = \varepsilon^{|\alpha|}(\partial^\alpha \psi)(\varepsilon \xi) = 0$ if $|\xi| \leq \frac{1}{\varepsilon}$ or $|\xi| \geq \frac{2}{\varepsilon}$, and $|\partial_\xi^\alpha(\psi(\varepsilon \xi))| \leq C_\alpha \varepsilon^{|\alpha|}$ for all $\xi \in \mathbb{R}^n$, where $C_\alpha = \sup_{\xi \in \mathbb{R}^n} |(\partial^\alpha \psi)(\xi)|$. If $\frac{1}{\varepsilon} \leq |\xi| \leq \frac{2}{\varepsilon}$, then $\varepsilon \leq \frac{2}{|\xi|} \leq \frac{4}{1+|\xi|}$. Hence, for any non-zero multi-index α , we have

$$|\partial_\xi^\alpha(\psi(\varepsilon \xi))| \leq C_\alpha 4^{|\alpha|} (1 + |\xi|)^{-|\alpha|} = C'_\alpha (1 + |\xi|)^{-|\alpha|}, \quad \xi \in \mathbb{R}^n$$

where $C'_\alpha = C_\alpha 4^{|\alpha|}$. Now, using Leibnitz's formula and the fact that $\sigma_j \in S^{m_j}$,

we can find positive constants $C_{\alpha,\gamma}$ and $C_{j,\beta,\gamma}$ such that $|D_\xi^\alpha D_x^\beta(\psi(\varepsilon\xi)\sigma_j(x,\xi))| :$

$$\begin{aligned}
&= |D_\xi^\alpha(\psi(\varepsilon\xi)(D_x^\beta\sigma_j)(x,\xi))| \\
&= \left| \sum_{\gamma \leq \alpha} \binom{\alpha}{\gamma} \left(D_\xi^{\alpha-\gamma}(\psi(\varepsilon\xi)) \right) \left(D_\xi^\gamma D_x^\beta\sigma_j \right)(x,\xi) \right| \\
&\leq \sum_{\gamma \leq \alpha} \binom{\alpha}{\gamma} C_{\alpha,\gamma} (1+|\xi|)^{-|\alpha|+|\gamma|} C_{j,\beta,\gamma} (1+|\xi|)^{m_j-|\gamma|} \\
&= \sum_{\gamma \leq \alpha} \binom{\alpha}{\gamma} C_{\alpha,\gamma} C_{j,\beta,\gamma} (1+|\xi|)^{m_j-|\alpha|} \\
&= C_{j,\alpha,\beta} (1+|\xi|)^{-1} (1+|\xi|)^{m_j+1-|\alpha|}
\end{aligned}$$

for all $x, \xi \in \mathbb{R}^n$, where $C_{j,\alpha,\beta} = \sum_{\gamma \leq \alpha} \binom{\alpha}{\gamma} C_{\alpha,\gamma} C_{j,\beta,\gamma}$. Now we choose ε_j such that

$$C_{j,\alpha,\beta} \varepsilon_j \leq 2^{-j}$$

for all multi-indices α and β such that $|\alpha + \beta| \leq j$. By the definition of ψ , we have

$$\psi(\varepsilon_j \xi) = 0$$

whenever $1 + |\xi| \leq \varepsilon_j^{-1}$. Hence,

$$\begin{aligned}
|D_\xi^\alpha D_x^\beta(\psi(\varepsilon_j \xi)\sigma_j(x,\xi))| &\leq 2^{-j} \varepsilon_j^{-1} (1+|\xi|)^{-1} (1+|\xi|)^{m_j+1-|\alpha|} \\
&\leq 2^{-j} (1+|\xi|)^{m_j+1-|\alpha|}
\end{aligned}$$

whenever $x, \xi \in \mathbb{R}^n$ and $|\alpha + \beta| \leq j$. Now, for any two multi-indices α_0 and β_0 , we take j_0 so large that $j_0 \geq |\alpha_0 + \beta_0|$ and $m_{j_0} + 1 \leq m_0$. Write

$$\begin{aligned}
\sigma(x, \xi) &= \sum_{j=0}^{j_0-1} \psi(\varepsilon_j \xi) \sigma_j(x, \xi) + \sum_{j=j_0}^{\infty} \psi(\varepsilon_j \xi) \sigma_j(x, \xi) \\
&= I(x, \xi) + J(x, \xi).
\end{aligned}$$

Since $I(x, \xi)$ is a finite sum, it follows that $I \in S^{m_0}$. Now

$$\begin{aligned}
|(D_\xi^{\alpha_0} D_x^{\beta_0} J)(x, \xi)| &\leq \sum_{j=j_0}^{\infty} \left| D_\xi^{\alpha_0} D_x^{\beta_0} (\psi(\varepsilon_j \xi) \sigma_j(x, \xi)) \right| \\
&\leq \sum_{j=j_0}^{\infty} 2^{-j} (1+|\xi|)^{m_j+1-|\alpha_0|} \\
&\leq \sum_{j=j_0}^{\infty} 2^{-j} (1+|\xi|)^{m_0-|\alpha_0|} \\
&= 2^{-j_0+1} (1+|\xi|)^{m_0-|\alpha_0|}.
\end{aligned}$$

So J is also in S^{m_0} i.e. $\sigma \in S^{m_0}$. We need to verify

$$\sigma - \sum_{j=0}^{N-1} \sigma_j \in S^{m_N}$$

To do so, we write

$$\begin{aligned} \sigma(x, \xi) - \sum_{j=0}^{N-1} \sigma_j(x, \xi) &= \sum_{j=0}^{\infty} \psi(\varepsilon_j \xi) \sigma_j(x, \xi) - \sum_{j=0}^{N-1} \sigma_j(x, \xi) \\ &= \sum_{j=0}^{N-1} (\psi(\varepsilon_j \xi) - 1) \sigma_j(x, \xi) + \sum_{j=N}^{\infty} \psi(\varepsilon_j \xi) \sigma_j(x, \xi). \end{aligned}$$

As before, we can show that $\sum_{j=N}^{\infty} \psi(\varepsilon_j \xi) \sigma_j(x, \xi) \in S^{m_N}$. Since $\psi(\varepsilon_j \xi) - 1 = 0$ for $j \leq N-1$ if $|\xi| \geq \frac{2}{\varepsilon_{N-1}}$, it follows from Proposition 1.2(5) that

$$\sum_{j=0}^{N-1} (\psi(\varepsilon_j \xi) - 1) \sigma_j(x, \xi) \in \bigcap_{m \in \mathbb{R}} S^m$$

and consequently $\sigma - \sum_{j=0}^{N-1} \sigma_j \in S^{m_N}$. Finally, if τ is another symbol such that $\tau \sim \sum_{j=0}^{\infty} \sigma_j$, then

$$\sigma - \tau = \left(\sigma - \sum_{j=0}^{N-1} \sigma_j \right) - \left(\tau - \sum_{j=0}^{N-1} \sigma_j \right) \in S^{m_N}$$

for every positive integer N . Since $m_N \rightarrow -\infty$ as $N \rightarrow \infty$, it follows that $\sigma - \tau \in \bigcap_{m \in \mathbb{R}} S^m$. This completes the proof of the theorem. $\square$

1.2 Partition of Unity

Theorem 1.10. There is a sequence $\{\varphi\}_{k=0}^{\infty}$ of functions in $C_0^{\infty}(\mathbb{R}^n)$ such that

- (i) $0 \leq \varphi_k(\xi) \leq 1$, $\xi \in \mathbb{R}^n$, $k = 0, 1, 2, \dots$,
- (ii) $\sum_{k=0}^{\infty} \varphi_k(\xi) = 1$, $\xi \in \mathbb{R}^n$,
- (iii) for each $\xi \in \mathbb{R}^n$, at least one and at most three of the φ_k 's are non-zero,
- (iv) $\text{supp}(\varphi_0) \subseteq \{\xi \in \mathbb{R}^n : |\xi| \leq 2\}$,
- (v) $\text{supp}(\varphi_k) \subseteq \{\xi \in \mathbb{R}^n : 2^{k-2} \leq |\xi| \leq 2^{k+1}\}$, $k = 1, 2, \dots$
- (vi) for each multi-index α , there is a constant $A_{\alpha} > 0$ such that

$$\sup_{\xi \in \mathbb{R}^n} |(\partial^{\alpha}(\varphi_k))(\xi)| \leq A_{\alpha} 2^{-k|\alpha|}, \quad k = 0, 1, 2, \dots$$

Proof. We pick ψ_0 be any function in $C_0^\infty(\mathbb{R}^n)$ such that $0 \leq \psi_0(\xi) \leq 1$ for all $\xi \in \mathbb{R}^n$, $\psi_0(\xi) = 1$ for $|\xi| \leq 1$ and $\psi_0(\xi) = 0$ for $|\xi| > 2$ such function exists by Proposition 1.8. Let ψ be any other function in $C_0^\infty(\mathbb{R}^n)$ such that $0 \leq \psi(\xi) \leq 1$ for all $\xi \in \mathbb{R}^n$, $\psi(\xi) = 1$ for $1 \leq |\xi| \leq 2$, and $\psi(\xi) = 0$ for $|\xi| < \frac{1}{2}$ or $|\xi| > 4$ ($\psi(\xi) = \psi_0(2^{-1}\xi) - \psi_0(2\xi)$). For $k = 1, 2, \dots$, define ψ_k by

$$\psi_k(\xi) = \psi\left(\frac{\xi}{2^{k-1}}\right), \quad \xi \in \mathbb{R}^n$$

and then define Ψ by

$$\Psi(\xi) = \sum_{k=0}^{\infty} \psi_k(\xi), \quad \xi \in \mathbb{R}^n$$

Obviously, we have

$$\text{supp}(\psi_0) \subseteq \{\xi \in \mathbb{R}^n : |\xi| \leq 2\}$$

and

$$\text{supp}(\psi_k) \subseteq \{\xi \in \mathbb{R}^n : 2^{k-2} \leq |\xi| \leq 2^{k+1}\}$$

for $k = 1, 2, \dots$. For each $\xi \in \mathbb{R}^n$, the series defining $\Psi(\xi)$ contains at most three non-zero consecutive terms. This implies that for each $\xi_0 \in \mathbb{R}^n$, there exists a neighbourhood U of ξ_0 and a positive integer N such that $\psi_k(\xi) = 0$ for all $\xi \in U$ and $k > N$. Hence $\Psi \in C^\infty(\mathbb{R}^n)$. It is easy to see that for each $\xi \in \mathbb{R}^n$, the series defining $\Psi(\xi)$ contains at least one non-zero term. Hence $\Psi(\xi) \neq 0$ for all $\xi \in \mathbb{R}^n$. Now, for $k = 0, 1, 2, \dots$, we define φ_k by

$$\varphi_k(\xi) = \frac{\psi_k(\xi)}{\Psi(\xi)}, \quad \xi \in \mathbb{R}^n$$

It is easy to see that $\varphi_k \in C_0^\infty$ for $k = 0, 1, 2, \dots$ and the sequence $\{\varphi_k\}_{k=0}^\infty$ satisfies the first five conditions. The last condition is trivially true for φ_0 . Hence, it remains to prove that the last condition is true for φ_k , $k = 1, 2, \dots$. To do this, we note that for each multi-index α , we have

$$\begin{aligned} (\partial^\alpha \varphi_k)(\xi) &= \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \left(\partial^\beta \left(\frac{1}{\Psi} \right) (\xi) \right) ((\partial^{\alpha-\beta} \psi_k)(\xi)) \\ &= \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \left(\partial^\beta \left(\frac{1}{\Psi} \right) (\xi) \right) \left((\partial^{\alpha-\beta} \psi_k) \left(\frac{\xi}{2^{k-1}} \right) 2^{-(k-1)|\alpha-\beta|} \right) \end{aligned}$$

Now, for each multi-index β ,

$$\partial^\beta \left(\frac{1}{\Psi} \right) = \sum_{\beta^{(1)} + \dots + \beta^{(l)} = \beta} C_{\beta^{(1)}, \dots, \beta^{(l)}} \frac{(\partial^{\beta^{(1)}} \Psi) \dots (\partial^{\beta^{(l)}} \Psi)}{\Psi^{l+1}},$$

where $C_{\beta^{(1)}, \dots, \beta^{(l)}}$ is a constant and the sum is taken over all possible multi-index $\beta^{(1)}, \dots, \beta^{(l)}$ which partition β . Suppose for every multi-index γ , there exists a constant $C_\gamma > 0$ such that

$$|(\partial^\gamma \Psi)(\xi)| \leq C_\gamma 2^{-k|\gamma|} \quad (*)$$

for all $\xi \in \text{supp}(\varphi_k)$ and $k = 0, 1, 2, \dots$. Then there exist positive constants $C_{\beta^{(1)}}, \dots, C_{\beta^{(l)}}$ such that

$$\begin{aligned} \left| \partial^\beta \left(\frac{1}{\Psi} \right) (\xi) \right| &\leq \sum_{\beta^{(1)} + \dots + \beta^{(l)} = \beta} |C_{\beta^{(1)}, \dots, \beta^{(l)}}| \frac{C_{\beta^{(1)}} \cdots C_{\beta^{(l)}} 2^{-k|\beta|}}{|\Psi(\xi)|^{l+1}} \\ &\leq C'_\beta 2^{-k|\beta|} \end{aligned}$$

for all $\xi \in \text{supp}(\varphi_k)$, where

$$C'_\beta = \sum_{\beta^{(1)} + \dots + \beta^{(l)} = \beta} |C_{\beta^{(1)}, \dots, \beta^{(l)}}| |C_{\beta^{(1)}} \cdots C_{\beta^{(l)}}|$$

There exists positive constants $C_{\alpha, \beta} > 0$ such that

$$\begin{aligned} |(\partial^\alpha \varphi_k)(\xi)| &\leq \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} C'_\beta 2^{-k|\beta|} C_{\alpha, \beta} 2^{-(k-1)|\alpha - \beta|} \\ &= A_\alpha 2^{-k|\alpha|} \end{aligned}$$

for all $k = 0, 1, 2, \dots$ and all $x \in \mathbb{R}^n$, where

$$A_\alpha = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} C'_\beta C_{\alpha, \beta} 2^{|\alpha - \beta|}.$$

Thus, it remains to prove (*), to do this we consider three cases:

- (i) Suppose $k = 0$. Then for all $\xi \in \text{supp}(\varphi_0)$, $\xi \in \text{supp}(\psi_j)$ for some $j = 0, 1, 2$ and hence

$$\Psi(\xi) = \psi_0(\xi) + \sum_{j=1}^2 \psi_j(\xi).$$

Therefore,

$$(\partial^\gamma \Psi)(\xi) = (\partial^\gamma \psi_0)(\xi) + \sum_{j=1}^2 (\partial^\gamma \psi_j) \left(\frac{\xi}{2^{j-1}} \right) 2^{-(j-1)|\gamma|}$$

for all $\xi \in \text{supp}(\varphi_0)$. Thus there is a constant $C_\gamma > 0$ such that

$$(\partial^\gamma \Psi)(\xi) \leq C_\gamma, \quad \xi \in \text{supp}(\varphi_0).$$

- (ii) Suppose $k = 1$. Then for all $\xi \in \text{supp}(\varphi_1)$, $\xi \in \text{supp}(\psi_j)$ for some $j = 0, 1, 2, 3$ and hence

$$\Psi(\xi) = \psi_0(\xi) + \sum_{j=1}^3 \psi_j(\xi).$$

Therefore,

$$(\partial^\gamma \Psi)(\xi) = (\partial^\gamma \psi_0)(\xi) + \sum_{j=1}^3 (\partial^\gamma \psi)(\frac{\xi}{2^{j-1}}) 2^{-(j-1)|\gamma|}$$

for all $\xi \in \text{supp}(\varphi_1)$. Thus there is a constant $C'_\gamma > 0$ such that

$$\begin{aligned} |\partial^\gamma(\Psi)(\xi)| &\leq C'_\gamma(2^{|\gamma|} + 1 + 2^{-|\gamma|} + 2^{-2|\gamma|}) \\ &= C'_\gamma(2^{2|\gamma|} + 2^{|\gamma|} + 1 + 2^{-|\gamma|})2^{-|\gamma|} \end{aligned}$$

for all $\xi \in \text{supp}(\varphi_1)$.

- (iii) Suppose $k \geq 2$. Then for all $\xi \in \text{supp}(\varphi_k)$, $\xi \in \text{supp}(\psi_j)$ for some $j = k-2, k-1, k, k+1, k+2$ and hence

$$\Psi(\xi) = \sum_{j=k-2}^{k+2} \psi_j(\xi).$$

Therefore

$$(\partial^\gamma \Psi)(\xi) = \sum_{j=k-2}^{k+2} (\partial^\gamma \psi)(\frac{\xi}{2^{j-1}}) 2^{-(j-1)|\gamma|}$$

for all $\xi \in \text{supp}(\varphi_k)$. Thus there is a constant $C''_\gamma > 0$ such that

$$\begin{aligned} |\partial^\gamma(\Psi)(\xi)| &\leq C''_\gamma[2^{-(k-3)|\gamma|} + 2^{-(k-2)|\gamma|} + 2^{-(k-1)|\gamma|} + 2^{-k|\gamma|} + 2^{-(k+1)|\gamma|}] \\ &= C''_\gamma[2^{3|\gamma|} + 2^{2|\gamma|} + 2^{|\gamma|} + 1 + 2^{-|\gamma|}]2^{-k|\gamma|} \end{aligned}$$

for all $\xi \in \text{supp}(\varphi_k)$. This completes the proof. □

Definition 1.11. Let $\sigma \in S^m$. For $k = 0, 1, 2, \dots$, we write

$$\sigma_k(x, \xi) = \sigma(x, \xi) \varphi_k(\xi)$$

for all $x, \xi \in \mathbb{R}^n$ and

$$K_k(x, z) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{iz \cdot \xi} \sigma_k(x, \xi) d\xi$$

for all $x, z \in \mathbb{R}^n$, where $\{\varphi_k\}$ is the partition of unity in Theorem 1.10.

Theorem 1.12. For all non-negative integers N , and multi-indices α and β , there exists a constant A , depending on m, n, N, α and β only, such that

$$\int_{\mathbb{R}^n} |z|^N |(\partial_x^\beta \partial_z^\alpha K_k)(x, z)| dz \leq A 2^{(m+|\alpha|-N)k}$$

for all $k = 0, 1, 2, \dots$.

Proof. Let γ be any multi-index, Plancherel's theorem: $\int_{\mathbb{R}^n} |z^\gamma (\partial_x^\beta \partial_z^\alpha K_k)(x, z)|^2 dz$

$$\begin{aligned} &= \int_{\mathbb{R}^n} |\partial_\xi^\gamma (\xi^\alpha (\partial_x^\beta \sigma_k)(x, \xi))|^2 d\xi \\ &= \int_{W_k} \left| \sum_{\gamma' \leq \gamma} \binom{\gamma}{\gamma'} \partial_\xi^{\gamma'} (\xi^\alpha (\partial_x^\beta \sigma)(x, \xi)) \left(\partial_\xi^{\gamma - \gamma'} \varphi_k \right) (\xi) \right|^2 d\xi \end{aligned}$$

where

$$W_0 = \{\xi \in \mathbb{R}^n : |\xi| \leq 2\}$$

and

$$W_k = \{\xi \in \mathbb{R}^n : 2^{k-2} \leq |\xi| \leq 2^{k+1}\}$$

for $k = 1, 2, \dots$. By part(vi) of Theorem 1.10 and the fact that $\xi^\alpha (\partial_x^\beta \sigma)$ is a symbol in $S^{m+|\alpha|}$, we get positive constants $C_{\alpha, \beta, \gamma'}$ and $C_{\gamma, \gamma'}$ such that

$$\begin{aligned} &\int_{\mathbb{R}^n} |z^\gamma (\partial_x^\beta \partial_z^\alpha K_k)(x, z)|^2 dz \\ &\leq \int_{W_k} \left| \sum_{\gamma' \leq \gamma} \binom{\gamma}{\gamma'} C_{\alpha, \beta, \gamma'} (1 + |\xi|)^{m+|\alpha|-|\gamma'|} C_{\gamma, \gamma'} 2^{-k|\gamma-\gamma'|} \right|^2 \\ &\leq \int_{W_k} \left| \sum_{\gamma' \leq \gamma} \binom{\gamma}{\gamma'} C_{\alpha, \beta, \gamma'} 2^{(2k+2)(m+|\alpha|-|\gamma'|)} C_{\gamma, \gamma'} 2^{-k|\gamma-\gamma'|} \right|^2 \\ &= 2^{k(2m+2|\alpha|-2|\gamma|)} \int_{W_k} \left| \sum_{\gamma' \leq \gamma} \binom{\gamma}{\gamma'} C_{\alpha, \beta, \gamma'} C_{\gamma, \gamma'} 2^{2(m+|\alpha|-|\gamma'|)} \right|^2 d\xi \\ &\leq 2^{k(n+2m+2|\alpha|-2|\gamma|)} C_n 2^n \left| \sum_{\gamma' \leq \gamma} \binom{\gamma}{\gamma'} C_{\alpha, \beta, \gamma'} C_{\gamma, \gamma'} 2^{2(m+|\alpha|-|\gamma'|)} \right|^2 \end{aligned}$$

where C_n is the constant with the property that the volume of the ball in $\mathbb{R}^n$ with radius r is equal to $C_n r^n$. Let $A_{\alpha, \beta, \gamma, m, n}$ be defined by

$$A_{\alpha, \beta, \gamma, m, n} = C_n 2^n \left| \sum_{\gamma' \leq \gamma} \binom{\gamma}{\gamma'} C_{\alpha, \beta, \gamma'} C_{\gamma, \gamma'} 2^{2(m+|\alpha|-|\gamma'|)} \right|^2$$

Then we have

$$\int_{\mathbb{R}^n} |z^\gamma (\partial_x^\beta \partial_z^\alpha K_k)(x, z)|^2 dz \leq A_{\alpha, \beta, \gamma, m, n} 2^{k(n+2m+2|\alpha|-2|\gamma|)}$$

Let N be any non-negative integer, then by Lemma 0.28 we have

$$\begin{aligned} \int_{\mathbb{R}^n} |z|^{2N} |(\partial_x^\beta \partial_z^\alpha K_k)(x, z)|^2 dz &\leq n^N \sum_{\gamma=N} \int_{\mathbb{R}^n} |z|^\gamma |(\partial_x^\beta \partial_z^\alpha K_k)(x, z)|^2 dz \\ &= A_1 2^{k(n+2m+2|\alpha|-2N)} \end{aligned}$$

for all $k = 0, 1, 2, \dots$, where

$$A_1 = n^N \sum_{|\gamma|=N} A_{\alpha, \beta, \gamma, m, n}.$$

By taking square roots, we get

$$\left(\int_{\mathbb{R}^n} |z|^{2N} |(\partial_x^\beta \partial_z^\alpha K_k)(x, z)|^2 dz \right)^{1/2} \leq A_2 2^{(\frac{n}{2}+m+|\alpha|-N)k}$$

for $k = 0, 1, 2, \dots$, where $A_2 = A_1^{1/2}$. Now write

$$\int_{\mathbb{R}^n} |z|^N |(\partial_x^\beta \partial_z^\alpha K_k)(x, z)| dz = \int_{|z| \leq 2^{-k}} + \int_{|z| > 2^{-k}}.$$

By Cauchy-Schwarz inequality, there exists a constant $A_3 > 0$, depending on m, n, N, α and β only, such that

$$\begin{aligned} \int_{|z| \leq 2^{-k}} &\leq \left(\int_{\mathbb{R}^n} |z|^{2N} |(\partial_x^\beta \partial_z^\alpha K_k)(x, z)|^2 dz \right)^{\frac{1}{2}} \left(\int_{|z| \leq 2^{-k}} dz \right)^{\frac{1}{2}} \\ &\leq A_3 2^{(\frac{n}{2}+m+|\alpha|-N)k} 2^{-\frac{nk}{2}} \\ &= A_3 2^{(m+|\alpha|-N)k} \end{aligned}$$

for $k = 0, 1, 2, \dots$. And by Cauchy-Schwarz inequality again, there is another $A_4 > 0$, depending on m, n, N, α and β only, such that

$$\begin{aligned} \int_{|z| > 2^{-k}} &\leq \left(\int_{\mathbb{R}^n} |z|^{2(N+n)} |(\partial_x^\beta \partial_z^\alpha K_k)(x, z)|^2 dz \right)^{\frac{1}{2}} \left(\int_{|z| > 2^{-k}} |z|^{-2n} dz \right)^{\frac{1}{2}} \\ &\leq A_4 2^{(\frac{n}{2}+m+|\alpha|-N-n)k} \left(\int_{S^{n-1}} \int_{2^{-k}}^\infty r^{-2n} r^{n-1} dr d\sigma \right)^{\frac{1}{2}} \end{aligned}$$

where $d\sigma$ is the surface measure on the unit sphere S^{n-1} . Thus,

$$\begin{aligned} \int_{|z| > 2^{-k}} &\leq A_4^{(\frac{n}{2}+m+|\alpha|-N-n)k} |S^{n-1}|^{\frac{1}{2}} 2^{\frac{nk}{2}} \\ &= A_5 2^{(m+|\alpha|-N)k} \end{aligned}$$

for $k = 0, 1, 2, \dots$ where $A_5 = A_4 |S^{n-1}|^{\frac{1}{2}}$ and $|S^{n-1}|$ is the surface area of S^{n-1} . And we are done. $\square$

Theorem 1.13. Let $f \in C^\infty(\mathbb{R}^n)$. Then for all positive integers N ,

$$f(\xi + \eta) = \sum_{|\alpha| < N} \frac{(\partial^\alpha f)(\xi)}{\alpha!} \eta^\alpha + N \sum_{|\gamma|=N} \frac{\eta^\gamma}{\gamma!} \int_0^1 (1-\theta)^{N-1} (\partial^\gamma f)(\xi + \theta\eta) d\theta$$

for all $\xi, \eta \in \mathbb{R}^n$.

Proof. The proof is by induction on N . For $N = 1$, we need to prove that

$$f(\xi + \eta) = f(\xi) + \sum_{|\gamma|=1} \frac{\eta^\gamma}{\gamma!} \int_0^1 (\partial^\gamma f)(\xi + \theta\eta) d\theta$$

for all $\xi, \eta \in \mathbb{R}^n$. To do this, we define a function $\varphi : \mathbb{R} \rightarrow \mathbb{C}$ by

$$\varphi(x) = f(\xi + x\eta), \quad x \in \mathbb{R}.$$

Then, by the fundamental theorem of calculus,

$$\varphi(1) = \varphi(0) + \int_0^1 \varphi'(t) dt.$$

Hence, we get the formula

$$f(\xi + \eta) = f(\xi) + \sum_{j=1}^n \int_0^1 (\partial_j f)(\xi + t\eta) \eta_j dt, \quad \xi, \eta \in \mathbb{R}^n$$

We now assume the statement is true for the positive integer N . Then, by the induction hypothesis, we have

$$\begin{aligned} f(\xi + \eta) &= \sum_{|\alpha| < N+1} \frac{(\partial^\alpha f)(\xi)}{\alpha!} \eta^\alpha \\ &= f(\xi + \eta) - \sum_{|\alpha| < N} \frac{(\partial^\alpha f)(\xi)}{\alpha!} \eta^\alpha - \sum_{|\alpha|=N} \frac{(\partial^\alpha f)(\xi)}{\alpha!} \eta^\alpha \\ &= N \sum_{|\gamma|=N} \frac{\eta^\gamma}{\gamma!} \int_0^1 (1-\theta)^{N-1} (\partial^\gamma f)(\xi + \theta\eta) d\theta - \sum_{|\alpha|=N} \frac{(\partial^\alpha f)(\xi)}{\alpha!} \eta^\alpha \end{aligned}$$

for all $\xi, \eta \in \mathbb{R}^n$. Now, using the fact that

$$N \int_0^1 (1-\theta)^{N-1} d\theta = 1$$

and the formula for $N = 1$, we have

$$f(\xi + \eta) = \sum_{|\alpha| < N+1} \frac{(\partial^\alpha f)(\xi)}{\alpha!} \eta^\alpha$$

$$\begin{aligned}
&= \sum_{|\gamma|=N} \frac{\eta^\gamma}{\gamma!} N \int_0^1 (1-\theta)^{N-1} ((\partial^\gamma f)(\xi + \theta\eta) - (\partial^\gamma f)(\xi)) d\theta \\
&= \sum_{|\gamma|=N} \frac{\eta^\gamma}{\gamma!} N \int_0^1 (1-\theta)^{N-1} \sum_{|\delta|=1} \frac{(\theta\eta)^\delta}{\delta!} \left(\int_0^1 (\partial^{\gamma+\delta} f)(\xi + \varphi\theta\eta) d\varphi \right) d\theta
\end{aligned}$$

Let $\rho = \varphi\theta$ we have

$$\begin{aligned}
&f(\xi + \eta) - \sum_{|\alpha| < N+1} \frac{(\partial^\alpha f)(\xi)}{\alpha!} \eta^\alpha \\
&= \sum_{|\gamma|=N} \sum_{|\delta|=1} \frac{\eta^{\gamma+\delta}}{\gamma!\delta!} N \int_0^1 (1-\theta)^{N-1} \left(\int_0^\theta (\partial^{\gamma+\delta} f)(\xi + \rho\eta) d\rho \right) d\theta.
\end{aligned}$$

By interchanging the order of integration we have

$$\begin{aligned}
&= \sum_{|\gamma|=N} \sum_{|\delta|=1} \frac{\eta^{\gamma+\delta}}{\gamma!\delta!} N \int_0^1 (\partial^{\gamma+\delta} f)(\xi + \rho\eta) \left(\int_\rho^1 (1-\theta)^{N-1} d\theta \right) d\rho \\
&= \sum_{|\gamma|=N} \sum_{|\delta|=1} \frac{\eta^{\gamma+\delta}}{\gamma!\delta!} \int_0^1 (\partial^{\gamma+\delta} f)(\xi + \rho\eta) (1-\rho)^N d\rho.
\end{aligned}$$

For all multi-indices γ and δ with $|\gamma| = N$ and $|\delta| = 1$, we have

$$\frac{(\gamma + \delta)!}{\gamma!\delta!} = \gamma \cdot \delta + 1$$

where $\gamma \cdot \delta$ is the inner product of γ and δ . Hence,

$$\begin{aligned}
&f(\xi + \eta) - \sum_{|\alpha| < N+1} \frac{(\partial^\alpha f)(\xi)}{\alpha!} \eta^\alpha \\
&= \sum_{|\gamma|=N} \sum_{|\delta|=1} \frac{\eta^{\gamma+\delta}}{(\gamma + \delta)!} (\gamma \cdot \delta + 1) \int_0^1 (\partial^{\gamma+\delta} f)(\xi + \rho\eta) (1-\rho)^N d\rho.
\end{aligned}$$

For each multi-index α of length $N+1$, we can write α in the form $\gamma + \delta$, where $|\gamma| = N$ and $|\delta| = 1$ by picking

$$\gamma = (\alpha_1, \alpha_2, \dots, \alpha_j - 1, \dots, \alpha_n)$$

and

$$\delta = (0, 0, \dots, 1, \dots, 0)$$

in which δ has 1 in the j^{th} position whenever $\alpha_j \geq 1$. Then we have

$$\gamma \cdot \delta = \alpha_j - 1$$

whenever $\alpha_j \geq 1$. Hence we have

$$\begin{aligned}
& f(\xi + \eta) - \sum_{|\alpha| < N+1} \frac{(\partial^\alpha f)(\xi)}{\alpha!} \eta^\alpha \\
&= \sum_{|\alpha|=N+1} \sum_{\alpha_j \geq 1} \frac{\eta^\alpha}{\alpha!} \alpha_j \int_0^1 (\partial^\alpha f)(\xi + \rho + \eta) (1 - \rho)^N d\rho \\
&= (N+1) \sum_{|\alpha|=N+1} \frac{\eta^\alpha}{\alpha!} \int_0^1 (\partial^\alpha f)(\xi + \rho + \eta) (1 - \rho)^N d\rho
\end{aligned}$$

Therefore, the statement is also true for the integer $N+1$ and we are done. $\square$

1.3 Products of Pseudo-Differential Operators

Theorem 1.14. Let $\sigma \in S^{m_1}$ and $\tau \in S^{m_2}$. Then the product $T_\sigma T_\tau$ of the pseudo-differential operators T_σ and T_τ is again a pseudo-differential operator T_λ , where λ is in $S^{m_1+m_2}$ and has the following asymptotic expansion:

$$\lambda \sim \sum_{\mu} \frac{(-i)^{|\mu|}}{\mu!} (\partial_\xi^\mu \sigma) (\partial_x^\mu \tau)$$

i.e.

$$\lambda - \sum_{\mu} \frac{(-i)^{|\mu|}}{\mu!} (\partial_\xi^\mu \sigma) (\partial_x^\mu \tau)$$

is a symbol in $S^{m_1+m_2-N}$ for every positive integer N .

Proof. For any function $\varphi \in \mathcal{S}$, we have

$$(T_{\sigma_k} \varphi)(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \sigma_k(x, \xi) \hat{\varphi}(\xi) d\xi$$

for all $x \in \mathbb{R}^n$, where $\sigma_k(x, \xi) = \sigma(x, \xi) \varphi_k(\xi)$ and $\{\varphi_k\}$ is the partition of unity in Theorem 1.10. Hence

$$\begin{aligned}
\sum_{k=0}^{\infty} (T_{\sigma_k} \varphi)(x) &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \hat{\varphi}(\xi) d\xi \\
&= (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \sigma(x, \xi) \hat{\varphi}(\xi) d\xi
\end{aligned}$$

for all $x \in \mathbb{R}^n$. The interchange of $\sum_{k=0}^{\infty}$ and $\int_{\mathbb{R}^n}$ can be justified by Fubini's theorem. Hence

$$T_\sigma \varphi = \sum_{k=0}^{\infty} T_{\sigma_k} \varphi$$

and the convergence of the series can be shown to be absolute and uniform for all $x \in \mathbb{R}^n$. Let $\varphi \in \mathcal{S}$, we first compute the symbol $T_{\sigma_k} T_\tau$, we have

$$\begin{aligned}
(T_{\sigma_k} T_\tau \varphi)(x) &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \sigma_k(x, \xi) \widehat{T_\tau \varphi}(\xi) d\xi \\
&= (2\pi)^{-n} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \sigma_k(x, \xi) \left(\int_{\mathbb{R}^n} e^{-i\xi \cdot y} (T_\tau \varphi)(y) dy \right) d\xi \\
&= (2\pi)^{-n} \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} e^{i(x-y) \cdot \xi} \sigma_k(x, \xi) d\xi \right) (T_\tau \varphi)(y) dy \\
&= (2\pi)^{-n/2} \int_{\mathbb{R}^n} K_k(x, x-y) (T_\tau \varphi)(y) dy
\end{aligned}$$

for all $x \in \mathbb{R}^n$. Hence, by the definition of a pseudo-differential operator and Fubini's theorem, the formula becomes $(T_{\sigma_k} T_\tau \varphi)(x)$

$$\begin{aligned}
&= (2\pi)^{-n} \int_{\mathbb{R}^n} K_k(x, x-y) \left(\int_{\mathbb{R}^n} e^{iy \cdot \eta} \tau(y, \eta) \hat{\varphi}(\eta) d\eta \right) dy \\
&= (2\pi)^{-n} \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} e^{-i(x-y) \cdot \eta} K_k(x, x-y) \tau(y, \eta) dy \right) \hat{\varphi}(\eta) d\eta \\
&= (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \eta} \lambda_k(x, \eta) \hat{\varphi}(\eta) d\eta
\end{aligned}$$

for all $x \in \mathbb{R}^n$, where

$$\lambda_k(x, \eta) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-i(x-y) \cdot \eta} K_k(x, x-y) \tau(y, \eta) dy$$

By a simple change of variable,

$$\lambda_k(x, \eta) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-iz \cdot \eta} K_k(x, z) \tau(x-z, \eta) dz$$

for all $x, \eta \in \mathbb{R}^n$. This suggests that

$$(T_\sigma T_\tau \varphi)(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{ix \cdot \eta} \lambda(x, \eta) \hat{\varphi}(\eta) d\eta$$

for all $x \in \mathbb{R}^n$, where

$$\lambda(x, \eta) = \sum_{k=0}^{\infty} \lambda_k(x, \eta)$$

for all $x, \eta \in \mathbb{R}^n$. Hence all we need to do is to show that $\lambda(x, \eta)$ is a symbol in $S^{m_1+m_2}$.

For $k = 0, 1, 2, \dots$, we define λ_k by

$$\lambda_k(x, \xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-iz \cdot \xi} K_k(x, z) \tau(x-z, \xi) dz$$

for all $x, \xi \in \mathbb{R}^n$. Now, by the Taylor's formula, we get

$$\tau(x - z, \xi) = \sum_{|\mu| < N_1} \frac{(-z)^\mu}{\mu!} (\partial_x^\mu \tau)(x, \xi) + R_{N_1}(x, z, \xi),$$

where

$$R_{N_1}(x, z, \xi) = N_1 \sum_{|\mu|=N_1} \frac{(-z)^\mu}{\mu!} \int_0^1 (1-\theta)^{N_1-1} (\partial_x^\mu \tau)(x - \theta z, \xi) d\theta.$$

for all $x, z, \xi \in \mathbb{R}^n$. By the Fourier inversion formula, we get

$$\lambda_k(x, \xi) = \sum_{|\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} (\partial_\xi^\mu \sigma_k)(x, \xi) (\partial_x^\mu \tau)(x, \xi) + T_{N_1}^{(k)}(x, \xi),$$

where

$$T_{N_1}^{(k)}(x, \xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-iz \cdot \xi} K_k(x, z) R_{N_1}(x, z, \xi) dz$$

for all $x, \xi \in \mathbb{R}^n$. For any positive integer N , the function $\lambda(x, \eta) = \sum_{k=0}^{\infty} \lambda_k(x, \eta)$ satisfies

$$\begin{aligned} & \lambda - \sum_{|\mu| < N} \frac{(-i)^{|\mu|}}{\mu!} (\partial_\xi^\mu \sigma)(\partial_x^\mu \tau) \\ &= \lambda - \sum_{|\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} (\partial_\xi^\mu \sigma)(\partial_x^\mu \tau) + \sum_{N \leq |\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} (\partial_\xi^\mu \sigma)(\partial_x^\mu \tau) \end{aligned}$$

where N_1 is any integer larger than N . Obviously,

$$\sum_{N \leq |\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} (\partial_\xi^\mu \sigma)(\partial_x^\mu \tau) \in S^{m_1+m_2-N}.$$

Hence, if we can prove that for all multi-indices α and β , there exists a constant $C_{\alpha, \beta} > 0$ such that

$$\left| (D_x^\alpha D_\xi^\beta \left(\lambda - \sum_{|\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} (\partial_\xi^\mu \sigma)(\partial_x^\mu \tau) \right))(x, \xi) \right| \leq C_{\alpha, \beta} (1 + |\xi|)^{m_1+m_2-N-|\beta|}$$

for all $x, \xi \in \mathbb{R}^n$, then we can conclude that $\lambda \in S^{m_1+m_2}$. We first make note that

$$\lambda - \sum_{|\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} (\partial_\xi^\mu \sigma)(\partial_x^\mu \tau) = \sum_{k=0}^{\infty} T_{N_1}^{(k)}.$$

Then, for any two multi-indices α and β , we need to estimate $D_x^\alpha D_\xi^\beta T_{N_1}^{(k)}$ for all $k = 0, 1, 2, \dots$. We need the following Lemma 1.15. Then, for all positive

integers N , and multi-indices α and β , we can choose a positive integer M such that

$$(1 + |\xi|^2)^{-M} (1 + |\xi|)^{m_2} \leq (1 + |\xi|)^{m_1 + m_2 - N - |\beta|}$$

for all $\xi \in \mathbb{R}^n$. With this M fixed, we can choose another positive integer N_1 so large that

$$m_1 + 2M - N_1 < 0.$$

Thus,

$$\begin{aligned} & \left| (D_x^\alpha D_\xi^\beta \left(\lambda - \sum_{|\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} (\partial_\xi^\mu \sigma)(\partial_x^\mu \tau) \right))(x, \xi) \right| \\ & \leq \sum_{k=0}^{\infty} C_{\alpha, \beta, M, N_1} (1 + |\xi|)^{m_1 + m_2 - N - |\beta|} 2^{(m_1 + 2M - N_1)k} \\ & = C_{\alpha, \beta} (1 + |\xi|)^{m_1 + m_2 - N - |\beta|} \end{aligned}$$

for all $x, \xi \in \mathbb{R}^n$, where

$$C_{\alpha, \beta} = C_{\alpha, \beta, M, N_1} \sum_{k=0}^{\infty} 2^{(m_1 + 2M - N_1)k}.$$

and we are done $\square$

Lemma 1.15. For all non-negative integers M , there exists a positive constant $C_{\alpha, \beta, M, N_1}$ such that

$$\left| \left(D_x^\alpha D_\xi^\beta T_{N_1}^{(k)} \right) (x, \xi) \right| \leq C_{\alpha, \beta, M, N_1} (1 + |\xi|^2)^{-M} (1 + |\xi|)^{m_2} 2^{(m_1 + 2M - N_1)k}$$

for all $x, \xi \in \mathbb{R}^n$ and $k = 0, 1, 2, \dots$.

Lemma 1.16. Let $R_{N_1}(x, z, \xi)$ be the function defined in the proof of Theorem 1.14. Then, for all multi-indices α, β and γ , there exists a constant $C_{\alpha, \beta, \gamma} > 0$ such that

$$\left| \left(\partial_z^\gamma \partial_x^\alpha \partial_\xi^\beta R_{N_1} \right) (x, z, \xi) \right| \leq C_{\alpha, \beta, \gamma} \left(\sum_{\gamma' \leq \gamma} |z|^{N_1 - |\gamma'|} \right) (1 + |\xi|)^{m_2 - |\beta|}$$

for all $x, z, \xi \in \mathbb{R}^n$.

Proof. We have

$$\left(\partial_x^\alpha \partial_\xi^\beta R_{N_1} \right) (x, z, \xi) = N_1 \sum_{|\mu| = N_1} \frac{(-z)^\mu}{\mu!} \int_0^1 (1 - \theta)^{N_1 - 1} (\partial_x^{\alpha + \mu} \partial_\xi^\beta \tau)(x - \theta z, \xi) d\theta$$

for all $x, z, \xi \in \mathbb{R}^n$. Hence, by Leibnitz's formula $\left(\partial_z^\gamma \partial_x^\alpha \partial_\xi^\beta R_{N_1}\right)(x, z, \xi)$

$$= N_1 \sum_{|\mu|=N_1} \sum_{\gamma' \leq \gamma} \binom{\gamma}{\gamma'} \frac{1}{\mu!} (\partial_z^{\gamma'} (-z)^\mu) \int_0^1 (1-\theta)^{N_1-1} \left(\partial_x^{\gamma-\gamma'+\alpha+\mu} \partial_\xi^\beta \tau\right)(x-\theta z, \xi) (-\theta)^{|\gamma-\gamma'|} d\theta$$

for all $x, z, \xi \in \mathbb{R}^n$. Use the fact that $\tau \in S^{m_2}$, there exist positive constants $C_{\gamma'}$ and $C_{\alpha, \beta, \gamma', \mu}$ such that $\left(\partial_z^\gamma \partial_x^\alpha \partial_\xi^\beta R_{N_1}\right)(x, z, \xi)$

$$\begin{aligned} &\leq N_1 \sum_{|\mu|=N_1} \sum_{\gamma' \leq \gamma} \binom{\gamma}{\gamma'} C_{\gamma'} |z|^{N_1-|\gamma'|} \int_0^1 C_{\alpha, \beta, \gamma', \mu} (1+|\xi|)^{m_2-|\beta|} d\theta \\ &\leq C_{\alpha, \beta, \gamma} \left(\sum_{\gamma' \leq \gamma} |z|^{N_1-|\gamma'|} \right) (1+|\xi|)^{m_2-|\beta|} \end{aligned}$$

for all $x, z, \xi \in \mathbb{R}^n$, where

$$C_{\alpha, \beta, \gamma} = N_1 \sum_{|\mu|=N_1} \sup_{\gamma' \leq \gamma} \left\{ \binom{\gamma}{\gamma'} C_{\gamma'} C_{\alpha, \beta, \gamma', \mu} \right\}$$

and we are done. $\square$

1.4 Elliptic Pseudo-Differential Operator

Definition 1.17. A symbol σ in S^m is said to be elliptic if there exist positive constants C and R such that

$$|\sigma(x, \xi)| \geq C(1+|\xi|)^m, \quad |\xi| \geq R$$

A pseudo-differential operator $Y = T_\sigma$ is said to be elliptic if its symbol is elliptic.

Theorem 1.18. Let σ be an elliptic symbol in S^m . Then there exists a symbol $\tau \in S^{-m}$ such that

$$\begin{aligned} T_\tau T_\sigma &= I + R \\ T_\sigma T_\tau &= I + S, \end{aligned}$$

where R and S are pseudo-differential operators with symbols in $\bigcap_{k \in \mathbb{R}} S^k$, and I is the identity operator.

Proof. The idea is to find a sequence of symbols $\tau_j \in S^{-m-j}$, $j = 0, 1, 2, \dots$. Let us assume that this has been done. Then, by Theorem 1.9, there exists symbol $\tau \in S^{-m}$ such that $\tau \sim \sum_{j=0}^\infty \tau_j$. Then by the product formula given in Theorem 1.14, the symbol λ of the product $T_\tau T_\sigma$ is in S^0 and satisfies the following relation:

$$\lambda - \sum_{|\gamma| < N} \frac{(-i)^{|\gamma|}}{\gamma!} (\partial_x^\gamma \sigma) (\partial_\xi^\gamma \tau) \in S^{-N}$$

for every positive integer N . But $\tau \sim \sum_{j=0}^{\infty} \tau_j$ implies that

$$\tau - \sum_{j=0}^{N-1} \tau_j \in S^{-m-N}$$

for every positive integer N . Hence

$$\lambda - \sum_{|\gamma| < N} \frac{(-i)^{|\gamma|}}{\gamma!} (\partial_x^\gamma \sigma) \sum_{j=0}^{N-1} (\partial_\xi^\gamma \tau_j) \in S^{-N}$$

for every positive integer N . But we can write

$$\begin{aligned} \sum_{|\gamma| < N} \frac{(-i)^{|\gamma|}}{\gamma!} \sum_{j=0}^{N-1} (\partial_\xi^\gamma \tau_j) (\partial_x^\gamma \sigma) &= \tau_0 \sigma + \sum_{l=1}^{N-1} \left(\tau_l \sigma + \sum_{\substack{|\gamma|+j=l \\ j < l}} \frac{(-i)^{|\gamma|}}{\gamma!} (\partial_\xi^\gamma \tau_j) (\partial_x^\gamma \sigma) \right) \\ &\quad + \sum_{\substack{|\gamma|+j \geq N \\ |\gamma| < N, j < N}} \frac{(-i)^{|\gamma|}}{\gamma!} (\partial_\xi^\gamma \tau_j) (\partial_x^\gamma \sigma) \end{aligned}$$

We choose τ_j , $j = 0, 1, 2, \dots$, in the following way: Define τ_0 by

$$\tau_0(x, \xi) = \begin{cases} \frac{\psi(\xi)}{\sigma(x, \xi)}, & |\xi| > R, \\ 0, & |\xi| \leq R \end{cases}$$

where ψ is any function in $C^\infty(\mathbb{R}^n)$ such that $\psi(\xi) = 1$ for $|\xi| \geq 2R$ and $\psi(\xi) = 0$ for $|\xi| \leq R$, and we define τ_l for $l \geq 1$ inductively by

$$\tau_l = - \left(\sum_{\substack{|\gamma|+j=l \\ j < l}} \frac{(-i)^{|\gamma|}}{\gamma!} (\partial_\xi^\gamma \tau_j) (\partial_x^\gamma \sigma) \right) \tau_0.$$

Then it can be checked easily that $\tau_j \in S^{-m-j}$, $j = 0, 1, 2, \dots$. Now, $\sigma(x, \xi) \tau_0(x, \xi) = 1$ for $|\xi| \geq 2R$. We see easily that $(\partial_\xi^\gamma \tau_j) (\partial_x^\gamma \sigma) \in S^{-N}$ whenever $|\gamma| + j \geq N$. By construction,

$$\sum_{|\gamma| < N} \frac{(-i)^{|\gamma|}}{\gamma!} \sum_{j=0}^{N-1} (\partial_\xi^\gamma \tau_j) (\partial_x^\gamma \sigma) - 1 \in S^{-N}$$

for every positive integer. Thus, $\lambda - 1 \in S^{-N}$ for every positive integer N . Hence, if we pick R to be the pseudo-differential operator with symbol $\lambda - 1$, we are done.

By a similar argument, we can find another symbol κ in S^{-m} such that

$$T_\sigma T_\kappa = I + R'$$

where R' is a pseudo-differential operator with symbol in $\bigcap_{k \in \mathbb{R}} S^k$. Then

$$T_\kappa + RT_\kappa = T_\tau + T_\tau R'$$

Since RT_κ and $T_\tau R'$ are pseudo-differential operators with symbols in $\bigcap_{k \in \mathbb{R}} S^k$, it follows that

$$T_\kappa = T_\tau + R''$$

where $R'' = T_\tau R' - RT_\kappa$ is another pseudo-differential operator with symbol in $\bigcap_{k \in \mathbb{R}} S^k$. Hence,

$$T_\sigma T_\tau = I + S$$

where $S = R' - T_\sigma R''$. Since S is a pseudo-differential operator with symbol in $\bigcap_{k \in \mathbb{R}} S^k$, this completes the proof. $\square$

Remark. If T_σ is an elliptic pseudo-differential operator, then it can be inverted modulo some error terms R and S with symbols in $\bigcap_{k \in \mathbb{R}} S^k$. In the theory of regularity of solutions of partial differential equations, operators with symbols in $\bigcap_{k \in \mathbb{R}} S^k$ are called infinitely smoothing and can be neglected.

Proposition 1.19. If T_σ is m_1 -elliptic and T_τ is m_2 -elliptic, then $T_\sigma T_\tau$ is $m_1 + m_2$ -elliptic.

Proof. By Theorem 1.14, the symbol $\lambda \in S^{m_1+m_2}$ for the composition $T_\sigma T_\tau$ satisfies

$$\lambda - \sigma\tau \in S^{m_1+m_2-1}.$$

So if we let $r = \lambda - \sigma\tau$, then there is a constant $D > 0$ such that

$$|r(x, \xi)| \leq D(1 + |\xi|)^{m_1+m_2-1}$$

Since σ, τ are m_1, m_2 -elliptic respectively, there are constants $C_1, C_2 > 0$ and $R > 0$ such that

$$\begin{aligned} |\sigma(x, \xi)| &\geq C_1(1 + |\xi|)^{m_1}, & |\xi| &\geq R \\ |\tau(x, \xi)| &\geq C_2(1 + |\xi|)^{m_2}, & |\xi| &\geq R. \end{aligned}$$

Hence,

$$\begin{aligned} |\lambda(x, \xi)| &= |\sigma(x, \xi)\tau(x, \xi) + r(x, \xi)| \\ &\geq |\sigma(x, \xi)||\tau(x, \xi)| - |r(x, \xi)| \\ &\geq C_1 C_2 (1 + |\xi|)^{m_1+m_2} - D(1 + |\xi|)^{m_1+m_2-1} \\ &\geq (C_1 C_2 - D(1 + |\xi|)^{-1})(1 + |\xi|)^{m_1+m_2}, \quad |\xi| \geq R \end{aligned}$$

Since $C_1, C_2, D > 0$ we can choose a constant $R' \geq R$ such that

$$C_1 C_2 - D(1 + R')^{-1} > 0.$$

Then

$$\begin{aligned} |\lambda(x, \xi)| &\geq (C_1 C_2 - D(1 + |\xi|)^{-1})(1 + |\xi|)^{m_1 + m_2} \\ &\geq (C_1 C_2 - D(1 + R')^{-1})(1 + |\xi|)^{m_1 + m_2}, \quad |\xi| \geq R' \end{aligned}$$

i.e. λ is $m_1 + m_2$ -elliptic. $\square$

Proposition 1.20. Suppose T_σ is an elliptic pseudo-differential operator and T_τ and T_κ are approximate inverse of T_σ . Then $\tau - \kappa \in \bigcap_{k \in \mathbb{R}} S^k$.

Proof. By the definition of an approximate inversion, there are pseudo-differential operators R and S with symbols in $\bigcap_{k \in \mathbb{R}} S^k$ such that

$$\begin{aligned} T_\sigma T_\tau &= I + R \\ T_\kappa T_\sigma &= I + S \end{aligned}$$

Therefore,

$$\begin{aligned} T_{\tau - \kappa} &= T_\tau - T_\kappa = IT_\tau - T_\kappa I \\ &= (T_\kappa T_\sigma - S)T_\tau - T_\kappa(T_\sigma T_\tau - R) \\ &= T_\kappa R - ST_\tau \end{aligned}$$

and $T_\kappa R - ST_\tau$ has is symbol in $\bigcap_{k \in \mathbb{R}} S^k$ by Theorem 1.14 and R, S have their symbols in $\bigcap_{k \in \mathbb{R}} S^k$. Hence, $\tau - \kappa \in \bigcap_{k \in \mathbb{R}} S^k$. $\square$

Proposition 1.21.

- (1) Let $P(x, D) = \sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha$, where the a_α 's are in $C^\infty(\mathbb{R}^n)$ and all their partial derivatives are bounded functions on $\mathbb{R}^n$. We call $\sum_{|\alpha|=m} a_\alpha(x) \xi^\alpha$ the principal symbol of $P(x, D)$ and denote it by $P_m(x, \xi)$. Then $P(x, D)$ is elliptic if and only if there exist positive constants C and R such that

$$|P_m(x, \xi)| \geq C(1 + |\xi|)^m, \quad |\xi| \geq R.$$

- (2) Let $P(D) = \sum_{|\alpha| \leq m} a_\alpha D^\alpha$ be a linear partial differential operator with constant coefficients. Let $P_m(\xi)$ be the principal symbol of $P(D)$. Then $P(D)$ is elliptic if and only if

$$P_m(\xi) = 0, \quad \xi \in \mathbb{R}^n \implies \xi = 0.$$

Proof.

- (1) ($\implies$): Assume $P(x, D)$ is elliptic. Its symbol is $P(x, \xi) = \sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha$ so there exist $C > 0$ and $R > 0$ such that

$$|P(x, \xi)| \geq C(1 + |\xi|)^m, \quad |\xi| \geq R.$$

Since $\sum_{|\alpha| \leq m-1} a_\alpha(x) \xi^\alpha \in S^{m-1}$, there exist $D > 0$ such that

$$\left| \sum_{|\alpha| \leq m-1} a_\alpha(x) \xi^\alpha \right| \leq D(1 + |\xi|)^{m-1}, \quad \xi \in \mathbb{R}^n.$$

Therefore,

$$\begin{aligned} |P_m(x, \xi)| &= \left| P(x, \xi) - \sum_{|\alpha| \leq m-1} a_\alpha \xi^\alpha \right| \\ &\geq |P(x, \xi)| - \left| \sum_{|\alpha| \leq m-1} a_\alpha \xi^\alpha \right| \\ &\geq \left(C - \frac{D}{1 + |\xi|} \right) (1 + |\xi|)^m, \quad |\xi| \geq R. \end{aligned}$$

Choose some $R' \geq R$ such that $C - \frac{D}{1+R'} > 0$ and let $C' = C - \frac{D}{1+R'}$. Then we have

$$|P_m(x, \xi)| \geq \left(C - \frac{D}{1 + |\xi|} \right) (1 + |\xi|)^m \geq C' (1 + |\xi|)^m, \quad |\xi| \geq R'$$

($\Leftarrow$) : Suppose there exist $C > 0$, $R > 0$ such that

$$|P_m(x, \xi)| \geq C(1 + |\xi|)^m, \quad |\xi| \geq R.$$

Then we have

$$\begin{aligned} |P(x, \xi)| &= \left| P_m(x, \xi) + \sum_{|\alpha| \leq m-1} a_\alpha(x) \xi^\alpha \right| \\ &\geq |P_m(x, \xi)| - \left| \sum_{|\alpha| \leq m-1} a_\alpha(x) \xi^\alpha \right| \\ &\geq \left(C - \frac{D}{1 + |\xi|} \right) (1 + |\xi|)^m, \quad |\xi| \geq R. \end{aligned}$$

i.e. $P(x, D)$ is elliptic.

(2) ($\Rightarrow$) : Assume there exist $C > 0$ and $R > 0$ such that

$$|P_m(\xi)| \geq C(1 + |\xi|)^m, \quad |\xi| \geq R.$$

In particular, $P_m(\xi) \neq 0$ for $|\xi| \geq R$. So if $\xi \neq 0$, then choosing $r > 0$ such that $|r\xi| \geq R$ implies that $P_m(r\xi) \neq 0$. But

$$P_m(r\xi) = \sum_{|\alpha|=m} a_\alpha(r\xi)^\alpha = \sum_{|\alpha|=m} a_\alpha r^{|\alpha|} \xi^\alpha = r^m \sum_{|\alpha|=m} a_\alpha \xi^\alpha = r^m P_m(\xi)$$

so $P_m(\xi) \neq 0$ as well. Therefore P_m has no non-zero roots. And 0 is clearly a root since $m \geq 1$ so P_m contains positive powers of ξ .

($\Leftarrow$) : Suppose 0 is the only root of $P_m(\xi)$. Since the unit sphere in $\mathbb{R}^n$ is compact, every continuous function has a minimum so we can let $C = \min_{|\xi|=1} |P(\xi)|$. As before, $P_m(r\xi) = r^m P_m(\xi)$ for all $r > 0$. So for all $|\xi| \geq 1$,

$$\begin{aligned} |P(\xi)| &= |\xi|^m \left| P\left(\frac{\xi}{|\xi|}\right) \right| \geq C |\xi|^m \\ &= C \frac{|\xi|^m}{(1 + |\xi|)^m} (1 + |\xi|)^m = \frac{C}{(1 + |\xi|^{-1})^m} (1 + |\xi|)^m \\ &\geq \frac{C}{2^m} (1 + |\xi|)^m \end{aligned}$$

so $P(x, D)$ is elliptic.

□

1.5 The Adjoint of Pseudo-Differential Operator

Definition 1.22. For any pair of functions φ and ψ in $\mathcal{S}$, we define (φ, ψ) by

$$(\varphi, \psi) = \int_{\mathbb{R}^n} \varphi(x) \overline{\psi(x)} dx.$$

Definition 1.23. Let σ be a symbol in S^m and T_σ its associated pseudo-differential operator. Suppose there exists a linear operator $T_\sigma^* : \mathcal{S} \rightarrow \mathcal{S}$ such that

$$(T_\sigma \varphi, \psi) = (\varphi, T_\sigma^* \psi), \quad \varphi, \psi \in \mathcal{S}$$

Then we call T_σ^* a formal adjoint of the operator T_σ .

Theorem 1.24. Let σ be a symbol in S^m . Then the formal adjoint of the pseudo-differential operator T_σ is again a pseudo-differential operator T_τ , where τ is a symbol in S^m and

$$\tau \sim \sum_{\mu} \frac{(-i)^{|\mu|}}{\mu!} \left(\partial_x^\mu \partial_\xi^\mu \bar{\sigma} \right) (x, \xi).$$

in other words,

$$\tau(x, \xi) - \sum_{|\mu| < N} \frac{(-i)^{|\mu|}}{\mu!} \left(\partial_x^\mu \partial_\xi^\mu \bar{\sigma} \right) (x, \xi)$$

is a symbol in S^{m-N} for every positive integer N .

Proof. For $k = 0, 1, 2, \dots$, we define σ_k and K_k by Definition 1.11. Then by the definition of a formal adjoint,

$$(T_{\sigma_k} \varphi, \psi) = (\varphi, T_{\sigma_k}^* \psi)$$

for all φ and ψ in $\mathcal{S}$. We have

$$\begin{aligned} (T_{\sigma_k} \varphi, \psi) &= \int_{\mathbb{R}^n} (T_{\sigma_k} \varphi) \overline{\psi(x)} dx \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} e^{ix \cdot \eta} \sigma_k(x, \eta) \hat{\varphi}(\eta) d\eta \right\} \overline{\psi(x)} dx \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} \hat{\sigma}_k(x, y - x) \varphi(y) dy \right\} \overline{\psi(x)} dx \end{aligned}$$

where

$$\hat{\sigma}_k(x, y) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-iy \cdot \eta} \sigma_k(x, \eta) d\eta$$

for all $x, y \in \mathbb{R}^n$. Therefore we have

$$\begin{aligned} (T_{\sigma_k} \varphi, \psi) &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} \hat{\sigma}_k(x, y - x) \overline{\psi(x)} dx \right\} \varphi(y) dy \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} K_k(x, x - y) \overline{\psi(x)} dx \right\} \varphi(y) dy \end{aligned}$$

for all $\varphi, \psi \in \mathcal{S}$. Therefore,

$$(T_{\sigma_k}^* \psi)(x) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} \overline{K_k(y, y - x)} \psi(y) dy$$

for all $x \in \mathbb{R}^n$. Hence, applying the Fourier inversion formula to the function ψ we have

$$\begin{aligned} (T_{\sigma_k}^* \psi)(x) &= (2\pi)^{-n} \int_{\mathbb{R}^n} \overline{K_k(y, y - x)} \left\{ \int_{\mathbb{R}^n} e^{iy \cdot \eta} \hat{\psi}(\eta) d\eta \right\} dy \\ &= (2\pi)^{-n} \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} e^{iy \cdot \eta} \overline{K_k(y, y - x)} dy \right\} \hat{\psi}(\eta) d\eta \\ &= (2\pi)^{-n} \int_{\mathbb{R}^n} e^{ix \cdot \eta} \left\{ \int_{\mathbb{R}^n} e^{i\eta \cdot (y - x)} \overline{K_k(y, y - x)} dy \right\} \hat{\psi}(\eta) d\eta \\ &= (2\pi)^{-n} \int_{\mathbb{R}^n} e^{ix \cdot \eta} \left\{ \int_{\mathbb{R}^n} e^{i\eta \cdot z} \overline{K_k(x + z, z)} dz \right\} \hat{\psi}(\eta) d\eta \end{aligned}$$

for all $x \in \mathbb{R}^n$. It is clear that $T_{\sigma_k}^* = T_{\tau_k}$ where

$$\tau_k(x, \eta) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{i\eta \cdot z} \overline{K_k(x + z, z)} dz.$$

Since $(T_\sigma \varphi, \psi) = \sum_{k=0}^{\infty} (T_{\sigma_k} \varphi, \psi)$ for all $\varphi, \psi \in \mathcal{S}$, it is clear that a good candidate for τ is given by

$$\tau(x, \eta) = \sum_{k=0}^{\infty} \tau_k(x, \eta)$$

for all $x, \eta \in \mathbb{R}^n$. Hence it remains to prove that τ is a symbol of order m and

$$(T_\sigma \varphi, \psi) = (\varphi, T_\tau \psi)$$

for all $\varphi, \psi \in \mathcal{S}$. Let N_1 be any positive integer, then by the Taylor's formula with integral remainder we have

$$K_k(x+z, z) = \sum_{|\mu| < N_1} \frac{z^\mu}{\mu!} (\partial_x^\mu K_k)(x, z) + R_{N_1}^{(k)}(x, z)$$

where $R_{N_1}^{(k)}(x, z)$

$$\begin{aligned} &= N_1 \sum_{|\mu|=N_1} \frac{z^\mu}{\mu!} \int_0^1 (1-\theta)^{N_1-1} (\partial_x^\mu K_k)(x + \theta z, z) d\theta \\ &= N_1 \sum_{|\mu|=N_1} \frac{z^\mu}{\mu!} \int_0^1 (1-\theta)^{N_1-1} (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{iz \cdot \xi} (\partial_x^\mu \sigma_k)(x + \theta z, \xi) d\xi d\theta \end{aligned}$$

Hence,

$$\tau_k(x, \eta) = \sum_{|\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} (\partial_x^\mu \partial_\eta^\mu \bar{\sigma}_k)(x, \eta) + T_{N_1}^{(k)}(x, \eta),$$

where

$$T_{N_1}^{(k)}(x, \eta) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{i\eta \cdot z} \overline{R_{N_1}^{(k)}(x, z)} dz$$

for all $x, \eta \in \mathbb{R}^n$. For any positive integer N ,

$$\begin{aligned} \tau &- \sum_{|\mu| < N} \frac{(-i)^{|\mu|}}{\mu!} \partial_x^\mu \partial_\eta^\mu \bar{\sigma} \\ &= \tau - \sum_{|\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} \partial_x^\mu \partial_\eta^\mu \bar{\sigma} + \sum_{N \leq |\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} \partial_x^\mu \partial_\eta^\mu \bar{\sigma} \end{aligned}$$

where N_1 is any integer larger than N . Obviously, $\sum_{N \leq |\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} \partial_x^\mu \partial_\eta^\mu \bar{\sigma}$ is in S^{m-N} . Hence, if we can prove that for all multi-indices α and β , there exists a constant $C_{\alpha, \beta} > 0$ such that

$$\left| \left(D_x^\alpha D_\eta^\beta \left\{ \tau - \sum_{|\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} \partial_x^\mu \partial_\eta^\mu \bar{\sigma} \right\} \right) (x, \eta) \right| \leq C_{\alpha, \beta} (1 + |\eta|)^{m-N-|\beta|}$$

for all $x, \eta \in \mathbb{R}^n$, then we can conclude that $\tau \in S^m$. To this end, we first make note that

$$\tau - \sum_{|\mu| < N_1} \frac{(-i)^{|\mu|}}{\mu!} \partial_x^\mu \partial_\eta^\mu \bar{\sigma} = \sum_{k=0}^{\infty} T_{N_1}^{(k)}.$$

Let α, β be any two multi-indices. Then

$$\begin{aligned} \left(D_x^\alpha D_\eta^\beta T_{N_1}^{(k)} \right) (x, \eta) &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{i\eta \cdot z} z^\beta \left(D_x^\alpha \overline{D_{N_1}^{(k)}} \right) (x, z) dz \\ &= (1 + |\eta|^2)^{-K} (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{i\eta \cdot z} (1 - \Delta_z)^K \left\{ z^\beta \left(D_x^\alpha \overline{R_{N_1}^{(k)}} \right) (x, z) \right\} dz \end{aligned}$$

where K is any positive integer. Let $P(D) = (1 - \Delta)^K$. Then we have

$$\begin{aligned} &(1 - \Delta_z)^K \left\{ z^\beta \left(D_x^\alpha \overline{R_{N_1}^{(k)}} \right) (x, z) \right\} \\ &= (1 - \Delta_z)^K N_1 \sum_{|\mu|=N_1} \frac{z^{\mu+\beta}}{\mu!} \int_0^1 (1 - \theta)^{N_1-1} (2\pi)^{-n/2} \\ &\quad + \int_{\mathbb{R}^n} (-i)^{|\alpha|} e^{-iz \cdot \xi} (\partial_x^{\alpha+\mu} \bar{\sigma}_k)(x + \theta z, \xi) d\xi d\theta \\ &= N_1 \sum_{|\mu|=N_1} \sum_{|\delta| \leq 2K} \frac{P^{(\delta)}(D)(z^{\mu+\beta})}{\mu! \delta!} \int_0^1 (1 - \theta)^{N_1-1} (2\pi)^{-n/2} \int_{\mathbb{R}^n} (\dots) d\xi d\theta, \end{aligned}$$

where

$$(\dots) = (-i)^{|\alpha|} \sum_{\rho \leq \delta} \binom{\delta}{\rho} \theta^{|\delta-\rho|} (D_z^\rho e^{-iz \cdot \xi}) (D_x^{\delta-\rho} \partial_x^{\alpha+\mu} \bar{\sigma}_k)(x + \theta z, \xi).$$

□

2 The Sobolev Spaces

Definition 2.1. For $-\infty < s < \infty$, we denote by J_s the pseudo-differential operator of which the symbol $\sigma_s(\xi)$ is given by

$$\sigma_s(\xi) = (1 + |\xi|^2)^{-s/2}, \quad \xi \in \mathbb{R}^n$$

The operator J_s is often called the Bessel potential of order s .

Proposition 2.2. We define $\sigma_s u$ for each $u \in \mathcal{S}'$ as the tempered distribution given by

$$(\sigma_s u)(\varphi) = u(\sigma_s \varphi), \quad \varphi \in \mathcal{S}$$

then $J_s u = \mathcal{F}^{-1} \sigma_s \mathcal{F} u$, for all $u \in \mathcal{S}'$.

Proof. Let $u \in \mathcal{S}'$, $\varphi \in \mathcal{S}$ then

$$\begin{aligned} \int_{\mathbb{R}^n} J_s u(x) \varphi(x) dx &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} e^{ix \cdot \xi} \sigma_s(\xi) \hat{u}(\xi) d\xi \right) \varphi(x) dx \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} \sigma_s(\xi) \hat{u}(\xi) \left(\int_{\mathbb{R}^n} e^{ix \cdot \xi} \varphi(x) dx \right) d\xi = \int_{\mathbb{R}^n} \sigma_s(\xi) \hat{u}(x) \check{\varphi}(\xi) d\xi \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} \sigma_s(\xi) \left(\int_{\mathbb{R}^n} e^{-i\xi \cdot x} u(x) dx \right) \check{\varphi}(\xi) d\xi \\ &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} e^{-i\xi \cdot x} \sigma_s(\xi) \check{\varphi}(\xi) d\xi \right) u(x) dx = \int_{\mathbb{R}^n} \widehat{\sigma_s \check{\varphi}}(x) u(x) dx. \end{aligned}$$

Thus,

$$J_s u(\varphi) = u(\widehat{\sigma_s \check{\varphi}}) = \mathcal{F} u(\sigma_s \check{\varphi}) = (\sigma_s \mathcal{F} u)(\check{\varphi}) = (\mathcal{F}^{-1}(\sigma_s \mathcal{F} u))(\varphi).$$

Therefore,

$$J_s u = \mathcal{F}^{-1}(\sigma_s \mathcal{F} u), \quad u \in \mathcal{S}'$$

□

Proposition 2.3. Let $u \in \mathcal{S}'$. Then

- (1) $J_s J_t u = J_{s+t} u$,
- (2) $J_0 u = u$.

Proof.

- (1) By Proposition 2.2, let $u \in \mathcal{S}'$

$$J_s J_t u = \mathcal{F}^{-1}(\sigma_s \mathcal{F}(\mathcal{F}^{-1}(\sigma_t \mathcal{F} u))) = \mathcal{F}^{-1}(\sigma_s(\sigma_t \mathcal{F} u)).$$

By definition of σ_s , it follows that $\sigma_s \sigma_t = \sigma_{s+t}$. For any $\varphi \in \mathcal{S}$,

$$(\sigma_s(\sigma_t \mathcal{F} u))(\varphi) = (\sigma_t \mathcal{F} u)(\sigma_s \varphi) = \mathcal{F} u(\sigma_t \sigma_s \varphi) = \mathcal{F} u(\sigma_{s+t} \varphi) = (\sigma_{s+t} \mathcal{F} u)(\varphi),$$

so $\sigma_s(\sigma_t \mathcal{F} u) = \sigma_{s+t} \mathcal{F} u$. It follows that

$$J_s J_t u = \mathcal{F}^{-1}(\sigma_{s+t} \mathcal{F} u) = J_{s+t} u$$

which completes the proof.

(2) Let $u \in \mathcal{S}'$, $J_0 u = \mathcal{F}^{-1} \mathcal{F} u = u$.

□

Definition 2.4. For $-\infty < s < \infty$ and $1 < p < \infty$, we define $H^{s,p}$ to be the set of all tempered distributions u for which $J_{-s}u$ is a function in $L^p(\mathbb{R}^n)$. It is obvious that $H^{s,p}$ is a vector space. It can be made into a normed vector space if we equip it with the norm $\|\cdot\|_{s,p}$, where

$$\|u\|_{s,p} = \|J_{-s}u\|_{L^p}, \quad u \in H^{s,p}.$$

We usually call $H^{s,p}$ the L^p -sobolev space of order s .

Theorem 2.5. $H^{s,p}$ is a Banach space with respect to the norm $\|\cdot\|_{s,p}$.

Proof. We need only prove completeness. To do this, let $\{u_k\}$ be a Cauchy sequence in $H^{s,p}$. Then, by the definition of $H^{s,p}$, the sequence $\{J_{-s}u_k\}$ is a Cauchy sequence in $L^p(\mathbb{R}^n)$. Since $L^p(\mathbb{R}^n)$ is complete, it follows that there exists a function $u \in L^p(\mathbb{R}^n)$ such that

$$J_{-s}u_k \rightarrow u \text{ in } L^p(\mathbb{R}^n)$$

as $k \rightarrow \infty$. Let $v = J_s u$ then $J_{-s}v = u$. Hence $J_{-s}v \in L^p(\mathbb{R}^n)$, i.e. $v \in H^{s,p}$. □

Proposition 2.6. J_t is an isometry of $H^{s,p}$ onto $H^{s+t,p}$. More precisely,

$$\|J_t u\|_{s+t,p} = \|u\|_{s,p}, \quad u \in H^{s,p}$$

Proof. Let $u \in H^{s,p}$. Then

$$\|J_t u\|_{s+t,p} = \|J_{-s-t} J_t u\|_{L^p} = \|J_{-s} u\|_{L^p} = \|u\|_{s,p}$$

Let $v \in H^{s+t,p}$. Then, $J_{-t}v \in H^{s,p}$ and $J_t J_{-t}v = v$. This proves that J_t is onto. □

Theorem 2.7 (Sobolev embedding theorem). Let $1 < p < \infty$ and $s \leq t$. Then $H^{t,p} \subseteq H^{s,p}$ and

$$\|u\|_{s,p} \leq \|u\|_{t,p}, \quad u \in H^{t,p}$$

Proposition 2.8. Let $s > 0$. If we define the function G_s on $\mathbb{R}^n$ by

$$G_s(x) := \frac{1}{2^{s/2} \Gamma(\frac{s}{2})} \int_0^\infty e^{-\frac{r}{2}} e^{-\frac{|x|^2}{2r}} r^{-(n-s)/2} \frac{dr}{r}, \quad x \in \mathbb{R}^n$$

then

- (i) $G_s \in L^1(\mathbb{R}^n)$,
- (ii) $\|G_s\|_{L^1} = (2\pi)^{n/2}$,
- (iii) $\hat{G}_s(\xi) = (1 + |\xi|^2)^{-s/2}$, $\xi \in \mathbb{R}^n$.

Proof. By Fubini's theorem,

$$\begin{aligned}
\int_{\mathbb{R}^n} |G_s(x)| dx &= \int_{\mathbb{R}^n} G_s(x) dx \\
&= \frac{1}{2^{s/2}\Gamma(\frac{s}{2})} \int_{\mathbb{R}^n} \left(\frac{1}{2^{s/2}\Gamma(\frac{s}{2})} \int_0^\infty e^{-\frac{r}{2}} e^{-\frac{|x|^2}{2r}} r^{-(n-s)/2} \frac{dr}{r} \right) dx \\
&= \frac{1}{2^{s/2}\Gamma(\frac{s}{2})} \int_0^\infty e^{-\frac{r}{2}} r^{-(n-s)/2} \left(\int_{\mathbb{R}^n} e^{-\frac{|x|^2}{2r}} dx \right) \frac{dr}{r} \\
&= \frac{(2\pi)^{n/2}}{2^{s/2}\Gamma(\frac{s}{2})} \int_0^\infty e^{-\frac{r}{2}} r^{\frac{s}{2}} \frac{dr}{r}
\end{aligned}$$

since

$$\int_{\mathbb{R}^n} e^{-\frac{|x|^2}{2r}} dx = (2\pi r)^{n/2}$$

But, for any $\varepsilon > 0$ and $a > 0$, we have

$$\varepsilon^{-a}\Gamma(a) = \int_0^\infty e^{-\varepsilon r} r^a \frac{dr}{r}.$$

Putting $\varepsilon = \frac{1}{2}$ and $a = \frac{s}{2}$, we get

$$\int_0^\infty e^{-\frac{r}{2}} r^{\frac{s}{2}} \frac{dr}{r} = 2^{\frac{s}{2}} \Gamma\left(\frac{s}{2}\right).$$

Hence we have

$$\int_{\mathbb{R}^n} |G_s(x)| dx = (2\pi)^{n/2}.$$

which proves (i) and (ii). Now, let $\varphi \in \mathcal{S}$, by Proposition 0.18,

$$\begin{aligned}
\int_{\mathbb{R}^n} \hat{G}_s(\xi) \varphi(\xi) d\xi &= \int_{\mathbb{R}^n} G_s(\xi) \hat{\varphi}(\xi) d\xi \\
&= \frac{1}{2^{s/2}\Gamma(\frac{s}{2})} \int_{\mathbb{R}^n} \left(\int_0^\infty e^{-\frac{r}{2}} e^{-\frac{|x|^2}{2r}} r^{-(n-s)/2} \frac{dr}{r} \right) \hat{\varphi}(\xi) d\xi
\end{aligned}$$

Using Fubini's theorem, we get

$$\begin{aligned}
&\int_{\mathbb{R}^n} \hat{G}_s(\xi) \varphi(\xi) d\xi \\
&= \frac{1}{2^{s/2}\Gamma(\frac{s}{2})} \int_0^\infty e^{-\frac{r}{2}} r^{-(n-s)/2} \left(\int_{\mathbb{R}^n} \hat{\varphi}(\xi) e^{-\frac{|\xi|^2}{2r}} d\xi \right) \frac{dr}{r}.
\end{aligned}$$

By Proposition 0.18 again, we get

$$\int_{\mathbb{R}^n} \hat{\varphi}(\xi) e^{-\frac{|\xi|^2}{2r}} d\xi = \int_{\mathbb{R}^n} \varphi(x) \hat{\psi}(x) dx,$$

where

$$\psi(x) = e^{-\frac{|x|^2}{2r}}, \quad x \in \mathbb{R}^n.$$

And we know that

$$\hat{\psi}(\xi) = r^{n/2} e^{-r|\xi|^2/2}, \quad \xi \in \mathbb{R}^n$$

Hence,

$$\int_{\mathbb{R}^n} \hat{\varphi}(\xi) e^{-\frac{|\xi|^2}{2r}} d\xi = r^{\frac{n}{2}} \int_{\mathbb{R}^n} \varphi(\xi) e^{-\frac{r|\xi|^2}{2}} d\xi.$$

Therefore,

$$\int_{\mathbb{R}^n} \hat{G}_s(\xi) \varphi(\xi) d\xi = \frac{1}{2^{s/2} \Gamma(\frac{s}{2})} \int_0^\infty e^{-\frac{r}{2}} r^{\frac{s}{2}} \left(\int_{\mathbb{R}^n} \varphi(\xi) e^{-\frac{r|\xi|^2}{2}} d\xi \right) \frac{dr}{r}.$$

Using Fubini's theorem again, we get

$$\begin{aligned} & \int_{\mathbb{R}^n} \hat{G}_s(\xi) \varphi(\xi) d\xi \\ &= \frac{1}{2^{s/2} \Gamma(\frac{s}{2})} \int_{\mathbb{R}^n} \varphi(\xi) \left(\int_0^\infty e^{-\frac{r}{2}} r^{\frac{s}{2}} e^{-\frac{r|\xi|^2}{2}} \frac{dr}{r} \right) d\xi \\ &= \frac{1}{2^{s/2} \Gamma(\frac{s}{2})} \int_{\mathbb{R}^n} \varphi(\xi) \left(\int_0^\infty e^{-\frac{r}{2}(1+|\xi|^2)} r^{\frac{s}{2}} \frac{dr}{r} \right) d\xi \end{aligned}$$

Putting $\varepsilon = (1 + |\xi|^2)/2$ and $a = \frac{s}{2}$ in $\varepsilon^{-a} \Gamma(a)$ we get

$$\begin{aligned} & \int_{\mathbb{R}^n} \hat{G}_s(\xi) \varphi(\xi) d\xi \\ &= \frac{1}{2^{s/2} \Gamma(\frac{s}{2})} \Gamma(\frac{s}{2}) 2^{s/2} \int_{\mathbb{R}^n} \varphi(\xi) (1 + |\xi|^2)^{-s/2} d\xi \\ &= \int_{\mathbb{R}^n} (1 + |\xi|^2)^{-s/2} \varphi(\xi) d\xi. \end{aligned}$$

And by Lemma 1.4, we can conclude that

$$\hat{G}_s(\xi) = (1 + |\xi|^2)^{-s/2}, \quad \xi \in \mathbb{R}^n.$$

□

Proposition 2.9. Let $s \geq 0$ and $1 \leq p < \infty$. Then

$$\|J_s u\|_{L^p} \leq \|u\|_{L^p}, \quad u \in L^p(\mathbb{R}^n).$$

Proof. Let $\varphi \in \mathcal{S}$. Then, by the definition of J_s ,

$$\widehat{J_s \varphi}(\xi) = (1 + |\xi|^2)^{-s/2} \hat{\varphi}(\xi), \quad \xi \in \mathbb{R}^n$$

On the other hand, Lemma 0.15

$$\begin{aligned} \widehat{G_s * \varphi}(\xi) &= (2\pi)^{n/2} \hat{G}_s(\xi) \hat{\varphi}(\xi) \\ &= (2\pi)^{n/2} (1 + |\xi|^2)^{-s/2} \hat{\varphi}(\xi), \quad \xi \in \mathbb{R}^n \end{aligned}$$

Hence, for all $\varphi \in \mathcal{S}$,

$$J_s \varphi = (2\pi)^{-n/2} (G_s * \varphi)$$

and

$$\|J_s \varphi\|_{L^p} \leq (2\pi)^{-n/2} \|G_s\|_{L^1} \|\varphi\|_{L^1} = \|\varphi\|_{L^1}$$

Since $\mathcal{S}$ is dense in $L^p(\mathbb{R}^n)$ for $1 \leq p < \infty$, it follows that J_s can be extended to a bounded linear operator on $L^p(\mathbb{R}^n)$. $\square$

Proof of Sobolev embedding theorem. Let $u \in H^{t,p}$. Then, by the definition of $H^{t,p}$, we have $J_{-t}u \in L^p(\mathbb{R}^n)$. Hence by Proposition 2.3, $J_{-s}u = J_{t-s}J_{-t}u$. By the definition of $H^{s,p}$ and Proposition 2.9, we get

$$\|u\|_{s,p} = \|J_{-s}u\|_{L^p} = \|J_{t-s}J_{-t}u\|_{L^p} \leq \|J_{-t}u\|_{L^p} = \|u\|_{t,p},$$

hence Theorem follows. $\square$

Theorem 2.10. For any symbol $\sigma \in S^m$, $T_\sigma : H^{m,p} \rightarrow H^{0,p}$ is a bounded linear operator for $1 < p < \infty$.

Theorem 2.11. Let σ be a symbol in S^m . Then $T_\sigma : H^{s,p} \rightarrow H^{s-m,p}$ is a bounded linear operator for $-\infty < s < \infty$ and $1 < p < \infty$.

3 Elliptic Regularity of PDEs

Definition 3.1. Let $P(x, D) = \sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha$ be a linear partial differential operator of order m such that

$$\sup_{x \in \mathbb{R}^n} |(D^\beta a_\alpha)(x)| < \infty, \quad |\alpha| \leq m$$

for all multi-indices β . Then $P(x, D)$ was a pseudo-differential operator with symbol $P(x, \xi)$ in S^m , where

$$P(x, \xi) = \sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha.$$

Theorem 3.2. let $P(x, D) = \sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha$ be a linear partial differential operator of order m . Let $x_0 \in \mathbb{R}^n$. Suppose that there exist positive constants C_1 and C_2 such that

$$\left| \sum_{|\alpha|=m} a_\alpha(x_0) \xi^\alpha \right| \geq C_1 |\xi|^m$$

and

$$\left| \sum_{|\alpha|=m} (a_\alpha(x) - a_\alpha(x_0)) \xi^\alpha \right| \leq C_2 |\xi|^m$$

for all x and $\xi \in \mathbb{R}^n$, where $C_2 < C_1$. If $u \in H^{s,p}$, $P(x, D)u = f$ and $f \in H^{s,p}$, then $u \in H^{s+m,p}$.

Lemma 3.3. Let $P(x, D) = \sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha$ be a linear partial differential operator of order m satisfying the hypothesis of Theorem 3.2. Then there exist positive constants C and R such that

$$|P(x, \xi)| \geq C(1 + |\xi|)^m, \quad |\xi| \geq R.$$

Proof. We begin by making note that

$$\begin{aligned} \left| \sum_{|\alpha|=m} a_\alpha(x) \xi^\alpha \right| &= \left| \sum_{|\alpha|=m} (a_\alpha(x) - a_\alpha(x_0)) \xi^\alpha + \sum_{|\alpha|=m} a_\alpha(x_0) \xi^\alpha \right| \\ &\geq \left| \sum_{|\alpha|=m} (a_\alpha(x) - a_\alpha(x_0)) \xi^\alpha \right| - \left| \sum_{|\alpha|=m} a_\alpha(x_0) \xi^\alpha \right| \\ &\geq (C_1 - C_2) |\xi|^m, \quad x, \xi \in \mathbb{R}^n \end{aligned}$$

then we can find positive constants C', C'' and R such that

$$\begin{aligned} \left| \sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha \right| &= \left| \sum_{|\alpha| < m} a_\alpha(x) \xi^\alpha + \sum_{|\alpha|=m} a_\alpha(x) \xi^\alpha \right| \\ &\geq \left| \sum_{|\alpha|=m} a_\alpha(x) \xi^\alpha \right| - \left| \sum_{|\alpha| < m} a_\alpha(x) \xi^\alpha \right| \\ &\geq C'(1 + |\xi|)^m - C''(1 + |\xi|)^{m-1} \\ &= (1 + |\xi|)^m (C' - C''(1 + |\xi|)^{-1}), \quad |\xi| \geq R. \end{aligned}$$

Obviously, we can find another positive constant $R_1 \geq R$ such that

$$C' - C''(1 + |\xi|)^{-1} \geq \frac{C'}{2}, \quad |\xi| \geq R_1.$$

Hence,

$$\left| \sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha \right| \geq \frac{C'}{2} (1 + |\xi|)^m, \quad |\xi| \geq R_1.$$

□

Proof of Theorem 3.2. By Lemma 3.3, $P(x, D)$ is an elliptic pseudo-differential operator with symbol $P(x, \xi)$ in S^m . Hence, by Theorem 1.18, we can find a symbol $\tau \in S^{-m}$ and a pseudo-differential operator R with symbol in $\bigcap_{k \in \mathbb{R}} S^k$ such that

$$T_\tau P(x, D) = I + R.$$

Hence, $u = T_\tau f - Ru$. Since $\tau \in S^{-m}$ and $f \in H^{s,p}$, it follows that $T_\tau f \in H^{s+m,p}$. Also $Ru \in H^{s+m,p}$ because R has symbol in S^{-m} and $u \in H^{s,p}$. Hence, $u \in H^{s+m,p}$. □

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